Climate Policy, Inequality, and The Role of Household Electrification

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Abstract

I study how household electrification shapes the distributional welfare effects of a carbon tax, focusing on the role of revenue recycling instruments. I develop a DSGE model with a discrete household electrification decision with financing frictions that slow adoption among low-income households, and non-homothetic energy demand. Using green subsidies as a recycling instrument lowers the peak tax rate required to achieve a given climate target, as it increases the share of renewable sources in electricity production and shifts energy demand towards electricity rather than fossil fuels. The lower tax rate and lower electricity price attenuate distortions in economic growth. However, as richer households electrify earlier, recycling carbon tax revenues through green subsidies disproportionately benefits higher-income households, causing inequality in welfare effects. Lump-sum transfers provide better insurance to financially constrained households but come at the cost of larger GDP and labor income losses. As an extension, I show that front-loaded carbon taxes reach climate targets with lower peak tax rates, limiting wage and GDP losses, but can amplify short-run inequality when combined with subsidy recycling.

Keywords: Green transition, Climate Policy, Carbon tax path, Inequality, Electrification

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1 Introduction

Climate policy is regarded as an important tool to reduce emissions and stop global warming. However it also raises distributional concerns as poorer households might be affected disproportionally financially. Movements such as the yellow-vests in France underscore the importance of careful winners and losers analyses for climate policy. This becomes even more relevant when climate policy design decisions, such as the instrument used to recycle revenues and the pace at which a carbon tax is phased in are main determinants of the distributional incidence of the tax. In this paper, I study the role of household electrification in shaping the distributional incidence of these policy choices. Household electrification entails a large upfront investment in a green technology, i.e. buying an electrical vehicle or heating system, and is unequally distributed over the income distribution. Poor households who lack the financial capacity to adopt green technologies might be disproportionally burdened by a carbon tax. However, the extent to which it does, and how this depends on policy design decisions remains quantitatively unknown.

For a given climate objective, e.g. a cap on total temperature increase, policymakers can choose to recycle the carbon tax revenues in several ways, such as directly redistributing it back to households, subsidizing green technology adoption by end-users or in the electricity sector, or use it as a financing tax for government expenditure. They can furthermore choose between a front-loaded or a more gradual carbon tax path. These decisions embed clear trade-offs, as more ambitious policy, subsidy recycling and front-loading, lowers the peak tax rate required to curb emissions and therefore lowers distortions, but are potentially regressive as poor households are not able to adapt to the policy via electrification due to the financing frictions.

Regarding the revenue recycling instrument, green subsidies encourage renewable electricity production or green technology adoption. Hence, they reduce emissions more for a given carbon tax rate compared to lump-sum redistribution. Therefore, given an exogenous temperature target, a lower tax rate suffices when subsidies are used, which in turn reduces GDP and wage declines caused by the tax. However, because subsidies lower the cost of electricity and green technology adoption, and because richer households adopt electricity-based green technologies earlier and more often, poorer households pay a disproportionate share of the carbon tax, while revenues mainly accrue to richer households. Furthermore, the smaller wage decline compared to lump-sum mostly benefits high productivity, rich households. A subsidy might thus be efficiency enhancing, but as these gains mainly appeal to the rich, they are likely to be regressive.

A similar trade-off holds for the duration of phasing in the tax. Front-loading the carbon tax curbs emissions earlier, such that a lower tax rate suffices for the same temperature target compared to back-loading the tax. This again attenuates efficiency distortions. However, as low income households with limited financial resources cannot adapt to green technologies to avoid paying carbon taxes, they remain more exposed to higher energy costs during the transition. Importantly, both the qualitative and quantitative extent of the trade-off might depend on the revenue recycling instrument used.

Household electrification plays a central role in the distributional incidence of climate policy because its adoption is highly unequal across the income distribution. A key feature of household electrification is that it

entails substantial upfront, investment costs. Empirical evidence from Danish near-universe register data, as documented by Berg et al. (2024) exploiting exogenous variation in unexpected inheritances, demonstrates that financial constraints constitute a key barrier to the adoption of electricity-based durables. Examining data from the Netherlands (2018) for homeowners, shows a clear positive correlation between income and the probability of early adoption of heat pumps, as can be seen in figure 13 in appendix A. This income gradient indicates that exposure to carbon pricing is systematically unequal along the income distribution. Low income households who lack the financial capacity to adapt to a green, electricity driven technology see a larger increase in energy prices, and therefore energy expenditures. This logic is reinforced by the fact that energy is a part necessary good. This creates a non-homotheticity in energy demand: poor households cannot substitute away from energy when relative prices rise.

Unequal rates of going green along the income and liquid wealth distribution are measured in empirical work in the Netherlands Dröes & Van Der Straten (2024) and for OECD countries Ameli & Brandt (2015). Furthermore, Kuhn & Schlattmann (2024) show that due to this heterogeneity, subsidizing the electrification process can induce transfers from poor to rich households. Crucitti et al. (2024) discuss how the adoption of green technologies such as heat pumps differs over the life-cycle. They find that as older households usually own bigger houses, they benefit more from proportional subsidies. A similar finding is found in Schlattmann (2024), where richer households own larger houses and therefore benefit more from a lower energy price. In this paper, I abstract from modeling a size complementarity and only focus on the extensive margin of electrification and examine how it affects the inequality versus inefficiency trade off in the policy decisions at hand.

In order to do so I build a quantitative life-cycle model with heterogeneous households, calibrated to the EU economy in 2015 (the Paris Agreement baseline year). Households are initially endowed with a brown durable but can choose to invest in a green durable enduring a large upfront investment cost. The durable is interpreted as a composite of heating and transport technologies (heat pump + EV for green versus gas boiler + gasoline car for brown). The two durables differ in fuel type used (electricity vs. fossil fuel) and energy efficiency, with the green durable more efficient.

On the production side, a representative firm combines a capital-labor composite with an energy aggregate to produce the final good. Energy is a CES bundle of electricity and fossil fuels, so firms adjust their input mix frictionless with relative prices. Electricity is produced from renewable and fossil-based technologies in a CES structure: renewable generation is capital-intensive, while fossil generation uses capital and fossil inputs. The electricity block is frictionless, so the renewable share responds contemporaneously to relative prices. Durables are produced from the final good at technology-specific productivities, implying durable prices equal inverse productivity.Initially, the green durable is more expensive, but its productivity converges over the transition to the productivity of the brown durable. The government implements carbon-tax paths and alternative revenue-recycling schemes subject to a balanced budget. I abstract from endogenous climate damages and impose an exogenous temperature objective, which allows a clean comparison of alternative tax-implementation horizons.

I find that more ambitious climate policy, with green subsidy recycling and front-loading significantly decreases the carbon tax rate required to limit global warming. With an electricity subsidy, GDP declines are about half the decrease compared to lump-sum transfer recycling. This result does not hold for the electrification subsidy, as direct household energy consumption has a small effect on total emissions in the calibrated economy.

The attenuated decline of GDP is caused by the significantly lower carbon tax, and hence reduced fossil fuel price increase and an electricity price decrease. The diminished distortions also cause the wage decline to be smaller with an electricity subsidy. Overall the lower electricity price and smaller wage decline benefit households, but do so disproportionally for households who are rich. As these households electrify earlier and more often, they enjoy the lower electricity price more. Furthermore, the proportional wage increase compared to lump-sum is larger for them in absolute terms.

Climate policy imposes the largest welfare losses on low-income, middle-aged households when carbon tax revenues are recycled through green subsidies. This pattern is driven by the electrification margin, which increases durable and energy expenditures for financially constrained households that are less able to adjust their technology. In contrast, lump-sum redistribution fully offsets the decline in lifetime consumption for low-income households. While this policy requires a higher peak carbon tax rate and therefore leads to larger long-run reductions in GDP, it eliminates the regressivity associated with subsidy recycling. Moreover, front-loading the carbon tax under lump-sum redistribution mitigates part of the GDP loss without imposing additional welfare costs on low-income households.

I contribute to the literature in two main ways. First, I construct a quantitative life-cycle model that jointly captures electricity decarbonization on the supply side and household electrification subject to financing frictions. Modeling both margins is crucial to identify the policy ordering: incorporating the supply side is needed to quantify how policy design affects the carbon tax required to meet a climate target, while electrification frictions determine how the resulting tax and price changes differentially affect poor households. This framework allows me to quantify the inequality–efficiency trade-off across policy instruments. Without a quantitative analyses, it is unclear whether the advantages brought upon by the lower carbon tax rate with green subsidies outperform the gain from the lump-sum transfer. Second, rich heterogeneity on the household side combined with a discrete electrification decision allows for an analysis of the life-cycle effects of climate policy. These two features together cause a heterogeneous impact of climate policy along the age distribution, as age is a co-determinant of the electrification probability. The second contribution lies in showing that climate policy has systematically different welfare effects across age cohorts.

The remainder of the paper is structured as follows. In section 1.1, I relate this paper to the existing literature on the green transition and inequality. In section 2, I discuss the model, that I parametrize in section 3. In section 4, I conduct policy experiments and analyze the results for equity and effectiveness. I conclude in section 5.

1.1 Related literature

A recent surge in interest has led to an increasing number of quantitative papers assessing inequality along the green transition. Benmir & Roman (2022) and Boehl et al. (2024) study how climate policy affects inequality in a setting where energy is modeled as a carbon-emitting input into the production process. In this setting, inequality arises from different channels such as changes in labor income, revenue redistribution or a decrease in future damages. More closely related to the framework developed in this paper is the literature that explicitly models energy as both a production input and a consumption good. Känzig (2023) examines the effect of carbon pricing in an economy where energy is modeled as a fossil fuel, and serves as an input in the production process and non-homothetically enters the utility function. A carbon tax affects households in two ways, indirectly via reduced economic activity and hence their income, and directly as the households consume energy. In a similar fashion, Fried et al. (2024) use a heterogeneous agent model with non-homothetic preferences for energy to examine the effect of a carbon tax with different rebate instruments on inequality. They find that rebating tax revenues via decreasing distortionary taxes and increasing the progressivity of labor taxes increases welfare more than lump-sum redistribution does.

Other papers make an explicit distinction between green and brown energy sources. In this set-up, climate policy reduces carbon emissions through decarbonization of the production of energy, i.e. electricity. In Ascari et al. (2025), the authors model two distinct energy sources, brown (fossil fuels) and green (renewable). A profit-maximizing energy firm aggregates these two sources of energy in a frictionless manner, based on relative prices and productivity, into a bundle of energy sold to every household and firm. A carbon tax raises the price of fossil fuels, such that the optimal mix contains more green energy. In their model opposed to my model, all households consume this same optimized bundle, such that the main inequality concern between rich and poor households stems from the non-homotheticity: the share of energy expenditures over income decreases in income. Closely related to their paper, is the paper by Hochmuth et al. (2025) which similarly models a non-homothetic demand for energy from households, and assesses the distributional effects of a carbon tax in an economy transitioning towards a carbon-neutral economy. In their paper too, all agents and firms consume the same bundle of energy. They find that both the elasticity of substitution between green and brown energy as well as the degree of non-homotheticity determine the quantitative degree of the inequality effects. In the current paper, I explicitly model both of these types of energy by adding an electrification margin for firms and households, rather than assuming that firms and all households consume the same energy mix. In this set-up, firms and households can consume both fossil fuels directly, and electricity as a composite of renewable and fossil generated electricity similarly modeled as energy in Ascari et al. (2025) and Hochmuth et al. (2025). The goal of climate policy is then to reduce emissions through both increased electrification and decarbonization of electricity production.

A similar distinction between types of energy is made in work examining the spatial margin in this decision by Labrousse & Perdereau (2024). The authors model two sources of energy, brown (fossil fuel) and green (electricity), that are both available for production and consumption. They include a household-specific CES optimized energy bundle based on locational choice (rural versus urban). This implies that a

household consumes its own optimal mix of brown and green energy, which is determined by the households' location, and the relative energy prices. Locational choice plays a key role in their electrification decision, as households in more rural areas have a stronger preference for fossil fuels than households in urban areas, and therefore are disproportionately hit by a carbon tax. Closely related to this, is the paper by Schlattmann (2024) which incorporates spatial heterogeneity in carbon demand, exploiting the fact that rural households emit more via direct energy demand. The paper analyzes the effect of a carbon tax on locational choice and energy demand. Contrary to the spatial dimension in these two papers, I exploit the fact that electrification requires a large upfront investment, rather than a moving decision. I therefore analyze a different mechanism behind the household decision to adjust to greener energy demand.

A paper that also examines this margin is the work by Kuhn & Schlattmann (2024). In their paper, they have a similar set up of modern versus old commitment goods, where the latter are thought of as more polluting. The authors examine the effect of subsidies for switching from the old to the modern commitment good on inequality. In a partial equilibrium setting, they analyze the effect of a subsidy that decreases the purchasing costs for modern commitment goods - implicitly defined as goods reliant on electricity rather than fossil fuels - with different financing instruments on inequality and transition efficiency. Similarly, Crucitti et al. (2024) focuses on heat pump uptake over the life-cycle in a setup where a large upfront investment is required. They analyze the effects of a proportional versus a lump-sum subsidy. These papers, however, refrain from modeling an electricity sector and thus only focus on the electrification decision. Similarly, in the paper by Fried (2026), green technology adoption serves as adaption against rising temperature variance caused by climate change rather than climate policy. Rich households can adapt to heat pumps that allow for cheaper regulation of the inside temperature in periods of extreme heat or cold. In this paper, I study the policy trade-off that arises from unequal green technology adoption rather than transfers from climate.

Lastly, this paper relates to the inter-generational comparison of climate policy burden. Fried et al. (2018) examine the distributional effects of different carbon tax revenue recycling instruments between generations by modeling a non-homothetic demand. They find that the generation that carries the biggest welfare burden of a carbon tax is highly conditional on the policy instrument used. They employ a steady state comparison, and evaluate how the future born generation is affected by the use of different revenue redistribution instruments. As opposed to their paper, I examine the short versus long-term effects over the transition, which are caused by the adoption friction rather than a steady state comparison.

2 Model

2.1 Households

Demographics Time is discrete. Households' age is indexed by $j = 1, 2, ..., J$. After period $J - 1$ the household dies with certainty. A household works for J_w periods, and then retires with certainty. The economy is populated by a continuum of households with mass one. At birth, households differ with respect to

their income $y_{i,0}$, initial wealth $a_{i,0}$ and the durable good they are born with, $\phi_{i,0}$. Households are indexed by i, however for notational convenience this index is dropped in most cases.

Preferences Households maximize expected life time utility denoted as

$$U = E\Big[\sum_{j=1}^{J-1} \beta^{j-1} u(c_j, s_j) + \beta^{J-1} w(b)\Big], \tag{1}$$

where β is the discount factor, $c > 0$ is consumption of the final consumption good and s is services derived from energy. Flow utility is a Stone-Geary utility function nested in a CRRA

$$u(c_j, s_j(\phi_j, e_j)) = \frac{1}{1-\sigma}(c_j^{\gamma}(s_j(\phi_j, e_j) - \bar{s})^{1-\gamma})^{1-\sigma}, \tag{2}$$

where γ denotes the households' relative taste for the final consumption good, $\frac{1}{\sigma}$ is the intertemporal elasticity of substitution and $\bar{s}$ is a subsistence level for services derived from energy. A subsistence level is included to capture the fact that energy demand is non-homothetic.[1]

The amount of services s, is a function of the level of energy consumption e and the type of durable good $\phi \in \{0, 1\}$ (brown vs. green), i.e. $d_\phi = \phi d_G + (1-\phi) d_B$. The type of the durable good, ϕ, determines the rate at which energy translates into services, such that

$$s(\phi_j, e_j) = \begin{cases} A_{s,G} e_j & : \text{if } \phi = 1 \implies d = d_G \\ A_{s,B} e_j & : \text{if } \phi = 0 \implies d = d_B. \end{cases} \tag{3}$$

Akin to the technologies described in Fried (2026), each durable good transforms energy (fossil fuels, electricity) into services of mobility and heating at a different rate, hence a higher productivity implies a more energy efficient durable good. The energy efficiency of the durable good is thus pinned down by its productivity parameters $A^{s,G}$ and $A^{s,B}$ for the green and brown durable good respectively[2].

A dying household receives utility from bequeathing its savings and a scrap value, φ, of its durable good. The utility derived from this is denoted as

$$w(b) = \eta \frac{(b + \underline{b})^{1-\sigma}}{1-\sigma} \tag{4}$$

with a strength of the bequest motive η, and where $\underline{b}$ denotes the extent to which bequeathing is a luxury good, similar to De Nardi (2004). Furthermore, the size of the bequest is defined as

$$b = a'_{J-1} + ((1 - \phi'_{J-1}) P^{d_B} + \phi'_{J-1} P^{d_G}). \tag{5}$$

[1] Most papers either use a subsistence level to capture this non-homotheticity Wöhrmüller (2024); Schlattmann (2024); Fried et al. (2024) or alternatively the flexible indirect utility function of Boppart (2014), adopted by among others Ascari et al. (2025); Hochmuth et al. (2025)

[2] These productivities are calibrated as the coefficient of performanceas defined by the International Energy Agency of the respective durable goods.

Durable good and energy As mentioned, there are two types of durable goods in the economy, a green, sustainable durable d_G and a brown, non-sustainable durable d_B. A durable good converts energy e into services (mobility, heating) at a rate specific to that durable good, $A_{s,G}$ or $A_{s,B}$[3]. Furthermore, the different types of durable good, require different types of energy. A green durable good is fueled with electricity, while a brown durable good requires fossil fuels. Hence, purchasing a green durable good is referred to as electrification.

Each household owns one type of durable good which is denoted using a binary state variable ϕ, such that the durable good owned by the household is given by $d_\phi = \phi d_G + (1-\phi)d_B$. At the start of its life, a household receives either a brown or green durable good from a warm-glow bequests, with a conditional probability based on its initial income level and determined by the dying cohorts' distribution. A household can electrify by selling its brown durable durable good ($\phi = 0$) for a scrap value φP^{d_B}, and buying a green durable good ($\phi' = 1$) for $P^{d_G}(1-s_{durable})$, where $s_{durable}$ is a government subsidy for electrifying households. A household is not allowed to switch from a green to a brown durable good, hence electrification is an absorbing state.

As said, the two goods d_G and d_B differ in the type of energy they require as fuel: the green durable good d_G relies on electricity (green energy), E with price P^E, whereas the brown durable good d_B relies on fossil fuels (brown energy), FF, with price P^{FF}. When the government levies a carbon tax τ_c, this increases the costs of fossil fuels. Household aggregate demand for electricity and fossil fuels are given as

$$E_H = \sum_{j=0}^{J-1} \int \phi_j e_j d\mu_j, \tag{6}$$

$$FF_H = \sum_{j=0}^{J-1} \int (1-\phi_j) e_j d\mu_j. \tag{7}$$

In case the household chooses to electrify, it can make use of a secured borrowing scheme to pay for the durable, denoted by m, subject to a down-payment constraint, i.e.

$$m' \leq \lambda^{LTV} P^G (1 - s_{durable}) \tag{8}$$

The household is required to repay the loan following an amortization schedule, which implies a payment of

$$\pi^m = \frac{(r+r_m)(1+(r+r_m))^{J-j}}{(1+(r+r_m))^{J-j}-1} m, \tag{9}$$

such that next periods loan is equal to $m' = (1+r^m)m - \pi^m$. The interest rate includes a premium r^m for borrowing by households.

A durable good requires maintenance, which is paid every period and is given by $\delta_\phi = \phi\delta_G P^G + (1-\phi)\delta_B P^B$.

Income and wealth Households supply their labor inelastically and receive income as an endowment

[3]This can be thought of as electricity or fossil fuels being translated into services of mobility (heating) by an electrical vehicle (heatpump) or gasoline vehicle (regular boiler)

each period. Labor income is denoted as $wY_{j,i}$, where $Y_{i,j}$ denotes labor productivity of a household i at age j, and is given by

$$log(Y_{j,i}) = \xi(j) + z_{j,i}. \tag{10}$$

such that labor productivity $Y_{j,i}$ consists of a deterministic life-cycle component $\chi(j)$ and an individual component, $z_{j,i}$, representing productivity differences between households. During the first J_w years, the idiosyncratic part of labor productivity $z_{j,i}$ of a household is stochastic and follows an AR(1) process,

$$z_{j,i} = \rho_y z_{j-1,i} + \epsilon_{j,i}. \tag{11}$$

After J_w periods, the household retires. After retirement, it deterministically receives a certain proportion f_{ret} of the income it received in the last period as a pension for the remaining $J - J_w$ periods.
Households can save in a risk-free asset denoted by a. They receive interest rate r on their savings every period. Unsecured borrowing is not allowed. Initial wealth is exogenously drawn from a mixture of a degenerate mass-point at zero wealth with probability p_0 and a univariate log-normal wealth distribution with variance $\sigma^2_{a_0}$ that integrates up to the savings bequeathed by dying households in that period.

2.1.1 Recursive formulation household problem

A household is born with initial wealth a_0, income y_0, and durable good ϕ_0. Thereafter, it solves the following problem during its lifetime, so for $j = 0, ..., J-1$, the household solves

$$V_j(a, \phi, m; Y, P) = \max\{V_j^{NoAdjust}(a, \phi, m; Y', P'), V_j^{Electrify}(a, \phi, m; Y', P') + \mu_{Adj}\} \tag{12}$$

Where $P = \{r, w, P^E, P^{FF}, P^G, P^B, \tau_c\}$ denotes the price vector. $\mu_{Adj} < 0$ is a negative preference shock, which represents a type-I extreme value shock. These shocks can be interpreted as a (temporary) aversion to change. It allows for the adjustment decision to be a probability rather than a discrete choice. The probability of adjustment is a function of the difference between $V_j^{Electrify}(a, \phi, m; Y', P') - V_j^{NoAdjust}(a, \phi, m; Y', P')$, but it spreads the mass which helps for calibration purposes. This helps in the calibration. Furthermore,

$$V_j^{NoAdjust}(a, \phi, m; Y, P) = \max_{a',c,e} u(c, s) + \beta E[V_{j+1}(a', \phi, m'; Y', P')] \tag{13}$$

subject to

$$wY + \mathcal{T} + (1 + r)a \geq c + a' + e[(1 - \phi)(P^{FF} + \tau_c) + \phi P^E] + \delta_\phi P^{d_\phi} + f(j)m \tag{14}$$

$$a' \geq 0 \tag{15}$$

$$f(j) = \frac{r_m(1 + r_m)^{J+1-j}}{(1 + r_m)^{J+1-j} - 1} \tag{16}$$

$$m' = (1 + r_m - f(j))m \tag{17}$$

And

$$V_j^{Electrify}(a,\phi,m;Y,P) = \max_{a',m',c,e} u(c,s) + \beta E[V_{j+1}(a',\phi'=1,m';Y',P')]. \tag{18}$$

subject to

$$wY + \mathcal{T} + (1+r)a + m' \geq c + a' + e\phi' P^E + \delta_\phi P^{d_\phi} + (P^{d_G}(1-s_{durable}) - \varphi P^{d_B}) \tag{19}$$

$$a' \geq 0, m' \leq \lambda P^{d_G}(1-s_{durable}) \tag{20}$$

Lastly $V_J = w(b)$ as in equation 4.

2.2 Production

There are three sectors on the production side of the economy, one for the final good Y^C, one for electricity Y^E and one for each durable good Y^{d_G} and Y^{d_B}. Fossil fuels are imported into the economy.

Final consumption good A representative firm produces output goods which can be used for production or converted in durable goods. Inputs are capital, labor and electricity with costs equal to interest rate plus depreciation, wage and the price of electricity. The technology is a constant returns to scale CES

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1-\psi)^{\frac{1}{\nu}}\left(A_{KC}K_C^{\alpha_c}L_C^{1-\alpha_c}\right)^{\frac{\nu-1}{\nu}} + \psi^{\frac{1}{\nu}}M^{\frac{\nu-1}{\nu}}\right]^{\frac{\nu}{\nu-1}} \tag{21}$$

where the energy composite M is given as

$$M = \left[\kappa^{\frac{1}{\iota}}(A_{EC}E_C)^{\frac{\iota-1}{\iota}} + (1-\kappa)^{\frac{1}{\iota}}(A_{FFC}FF_C)^{\frac{\iota-1}{\iota}}\right]^{\frac{\iota}{\iota-1}} \tag{22}$$

Electricity There is an electricity producer that inputs capital and fossil fuels and converts it into electricity.

The profits of the electricity firm are given as follows by an aggregator of renewable and fossil generated electricity multiplied by the price of electricity, minus the costs for inputs, hence

$$\Pi(K_{E_R}, K_{E_F}, FF_E) = P_E\big(\alpha_R^{\frac{1}{\lambda}}(Y^{E_R}(K_{E_R}))^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}(Y^{E_F}(K_{E_F}, FF_E)^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} \tag{23}$$

$$- (r+\delta)(1-s_{GE})K_{E_R} - (r+\delta)K_{E_F} - (P^{FF}+\tau_c)FF_E \tag{24}$$

Where s_{GE} denotes the subsidy for the cost of green electricity generation. The production technologies for renewable and fossil generated electricity are

$$Y^{E_R}(K_{E_R}) = A_{KR}K_{E_R} \tag{25}$$

$$Y^{E_F}(K_{E_F}, FF_E) = [\zeta^{\frac{1}{\omega}}(A_{KF}K_{E_F})^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}}FF_E^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}} \tag{26}$$

Durable goods In this economy, the durable goods can be produced by converting the consumption good (with price 1, as it is the numeraire) at some rate A_G, A_B, such that production functions are given as

$$Y^{d_G}(C_G) = A_G C_G \tag{27}$$

$$Y^{d_B}(C_B) = A_B C_B \tag{28}$$

These imply input demand functions

$$C_G = \frac{1}{A_G} Y^{d_G} \tag{29}$$

$$C_B = \frac{1}{A_B} Y^{d_B} \tag{30}$$

and prices $P^G = \frac{1}{A_G}$, $P^B = \frac{1}{A_B}$.

2.3 Government

The government runs a balanced budget. It receives a carbon tax every period, and gives out a lump-sum uniform transfer, a subsidy for capital costs to generate renewable energy generation or a subsidy for the costs of electrification, such that

$$G + \mathcal{T} + s_{E_R} K_{E_R}(r+\delta) + s_{durable} P^G \sum_{j=0}^{J-1} \int (\phi' - \phi) d\mu_j = \tau_c [FF_H + FF_E + FF_C] \tag{31}$$

2.4 Equilibrium

Given an exogenous fossil fuel price P^{FF} and a carbon tax τ_c, equilibrium is defined by a price vector $P : \{P^{FF}, \tau_c, r, w, P^E, P^G, P^D\}$ and distribution μ over state variables (a, ϕ, m, y, j) such that

1. Given prices, households maximize utility giving rise to value functions $V_j(a, \phi; y, P, \mathcal{T})$, $V_j^{NA}(a, \phi; y, P, \mathcal{T})$, $V_j^{A}(a, \phi; y, P, \mathcal{T})$ and policy functions $a'_j(a, \phi, m; y, P, \mathcal{T})$, $c_j(a, \phi, m; y, P, \mathcal{T})$, $e_j(a, \phi, m; y, P, \mathcal{T})$, $\phi'_j(a, \phi, m; y, P, \mathcal{T})$ and $m'_j(a, \phi, m; y, P, \mathcal{T})$

2. The final-output good firm, electricity firm, and the durable good firms maximize profits

3. All financial wealth bequeathed and all durables bequeathed equals the initial, aggregate wealth of the new-born generation, that is, for wealth

$$\int a'(\mathbf{x}) d\mu_{J-1} = \int a d\mu_{-1}, \tag{32}$$

   and durable goods are passed on conditional on initial income level, such that the joint distribution of durables and income is similar in the dying and newborn generation

4. The government budget constraint holds

$$G + \mathcal{T} + s_{E_R} K_{E_R}(r+\delta) + s_{durable} P^G \sum_{j=0}^{J-1} \int (\phi' - \phi) d\mu_j = \tau_c [\sum_{j=0}^{J-1} \int e(1-\phi') d\mu_j + FF_E + FF_C] \tag{33}$$

5. Prices are such that markets clear for

$$\text{Capital:} \quad \sum_{j=0}^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \tag{34}$$

$$\text{Labor:} \quad \sum_{j=0}^{J-1} \int y d\mu_j = L_C \tag{35}$$

$$\text{Gr. Durable:} \quad Y^{D_G} = \sum_{j=0}^{J-1} \int \phi \delta_G + (\phi' - \phi) d\mu_j \tag{36}$$

$$\text{Br. Durable:} \quad Y^{D_B} = \sum_{j=0}^{J-1} \int (1-\phi)\delta_B - \varphi(\phi' - \phi) d\mu_j \tag{37}$$

$$\text{Electricity:} \quad Y^E = \sum_{j=0}^{J-1} \int e\phi' d\mu_j + E_C \tag{38}$$

and the aggregate resource constraint clears by Walras law.

3 Parametrization baseline model

In order to assess the quantitative trade-offs behind climate policy during the green transition, I parametrize the economy to match the initial steady state to the pre-transition period. I focus on the European green transition, and calibrate to the year 2015.

Preferences Households are born at age 25, work until 65 and die at 75. This implies J is 50, and J_w is 40. All generations are equally sized, hence the size of a generation equals $N = 1/J$. I set the intertemporal elasticity of substitution $1/\sigma$ to $1/2$, which is a standard value in the lifecycle literature. The remaining preferences parameters - $\beta, \gamma, \bar{s}, \eta, \underline{b}$ - are calibrated internally using a simulated method of moments. The key moments targeted in order to discipline these parameters are the mean savings in the economy, the average energy expenditure over total expenditure, the degree of non-homothetecity in energy expenditures, the change in saving holdings during the retirement phase and inequality in bequests. This is discussed in more detail below.

Parameter	Description	Value	Internal	Target/source
Households				
Preferences				
β	Discount factor	0.941	yes	See table 2
$1/\sigma$	Intertemporal substitution	1/2	no	Standard value
γ	Taste for consumption	0.917	yes	See table 2
$\bar{s}$	Energy service subsistence level	7.6	yes	See table 2
η	Strength of bequest motive	31.49	yes	See table 2
$\underline{b}$	Luxury of bequest	3.27	yes	See table 2
Durable goods and energy				
$A_{s,G}$	Energy efficiency d_G	1.82	no	See text
$A_{s,B}$	Energy efficiency d_B	0.47	no	See text
δ_G, δ_B	Depreciation	0.02, 0.01	no	Schloter (2022)
φ	Resale value	0.27	no	Gilmore & Lave (2013)
$1 - \lambda_{LTV}$	Down payment requirement	0.1	no	Own assumption
r_m	Loan premium	0.03	no	–
μ_{Adj}	Location preference shock	-1.2	yes	See table 2
σ_{Adj}	Scale preference shock	0.32	yes	See table 2
Income and wealth				
ρ_y	Persistence	0.98	no	DNB Household Survey
σ_ϵ	Std. dev. income	0.1	no	DNB Household Survey
σ_{a_0}	Std. dev. wealth at age 0	0.556	no	DNB Household Survey
p_0	Zero-wealth share at age 0	0.39	no	DNB Household Survey
Production				
Electricity				
α_R	Renewable share	0.5	no	Hochmuth et al. (2025)
λ	EoS green and brown electricity	1.8	no	Papageorgiou et al. (2017)
ζ	Capital share E_F	0.3	no	Own assumption
ω	EoS capital and fossil fuels in E_F	0.25	no	Coenen et al. (2024)
A_{KR}	Productivity green capital	14.72	yes	See table 2
A_{KF}	Productivity brown capital	17.46	yes	See table 2
Durable goods				
A_G	Productivity Y^{d_G}	0.86	no	Kuhn & Schlattmann (2024)
A_B	Productivity Y^{d_B}	2.2	no	Price of d_B
Final consumption good				
α_c	Capital share	0.33	no	Literature
ν	EoS capital-labor and energy	0.2	no	Realistically low
ψ	Capital-labor share	0.1	no	Hochmuth et al. (2025)
ι	EoS Electricity and Fossil Fuel	1.5	no	Papageorgiou et al. (2017)
κ	Electricity share	0.2	no	Papageorgiou et al. (2017)
A_{KC}	Productivity capital	1.092	yes	Normalization w
A_{EC}	Productivity electricity	0.015	yes	See table 2
A_{FFC}	Productivity fossil fuels	0.00048	yes	See table 2

Table 1: Parameter values

Durable good and Energy The composite durable converts energy into heat and mobility services. I follow Kuhn & Schlattmann (2024) in defining 2/3 of the composite good as the car and the remaining part as the heating system, according to expenditures on energy for both components.

The rate at which energy is translated into services, $A^{s,G}$ and $A^{s,B}$ for the green and brown durable good respectively are defined as energy efficiency. I calibrate the efficiency levels of the two technologies separately for heat and for mobility and then aggregate them for the composite good.

Energy efficiency is defined as the rate at which a unit of energy, either fossil fuel or electricity, translates into useful energy by the technology at hand. That is, how much useful energy is extracted from using the primary (or secondary for electricity) source of energy. Due to transformation losses or gains at the time of usage, energy services consumed is not the same as final energy demand.

To start with heating, I follow energy efficiency estimates from IEA (2022)[4]. They define the concept of useful energy as energy available to satisfy the needs of the user, and refer to this as energy services demand, similar to my model. In their report, they refer to the difference between useful energy and energy used to run the technology as the coefficient of performance (COP), which directly translates into $A_H^{s,G}$ and $A_H^{s,B}$ in my model. They state that heat pumps have a COP of around 4, while there is a huge range from industrial heat pumps having a COP of 3 and district heating networks with large heat pumps showing a COP of 8. I choose to set $A_H^{s,G}$ to 4. Natural gas boilers on the other hand exhibit a COP of around 0.8, hence I set $A_H^{s,B}$ to 0.8.

For mobility, I follow estimates from the US department of energy from 2024[5]. They find that for electrical vehicles the energy conversion rate from energy input to useful energy, $A_M^{s,G}$ lies around 0.75[6]. For a gasoline car energy efficiency $A_M^{s,B}$ is lower, as it only translates 0.3 of total energy into useful energy and the remainder is lost as heat. This adds up to 1.82 for $A^{s,G}$, and 0.47 for $A^{s,B}$.

For other parameters, I follow the empirical literature. Following estimates from Schloter (2022), electrical vehicles depreciate faster than gasoline vehicles. Estimates for heat pumps are diverging. I set the depreciation for the green durable good to 2% as opposed to 1% for the brown durable good. Regarding the resale value, I follow a resale value for cars from Gilmore & Lave (2013) of 40% and 0% for heating systems. For the composite good, this implies a resale value of 27%.

The price of fossil fuels is calculated as a composite of the gasoline and the natural gas price, for mobility and heating purposes, converted to kWh. To start with the price of natural gas, from Eurostat[7], the price for one kWh of natural gas is 0.05 euros in Europe, 2015. From the weekly oil bulletin, the price of gasoline (Euro-super95) is 529.7 euros per 1000 liter. Taking as a conversion 9.3 kWh per liter of gasoline, the price of gasoline per kWh in the Europe, 2015 is 0.0569 euros. As the composite good consists of 67% mobility purposes, the price of fossil fuels would be 0.0546 euros per kWh. The model unit is defined as 1000 kWh,

[4]The Future of Heatpumps report

[5]https://www.energy.gov/eere/vehicles/articles/fotw-1360-sept-16-2024-typical-ev-87-91-efficient-compared-30-conventional

[6]taking into account cars with and without regenerative breaking

[7]Online data code: nrg_pc_202

and normalized by dividing by mean income, such that the price in the model is set to 0.0023. [8]

With respect to the parameters governing the characteristics of secured borrowing for the purchase of a durable good. The minimum down-payment requirement is set to 10% by own assumption in order to set it realistically. Lastly, I set a premium on loans for households to match the fact that these loans often require high interest payments.

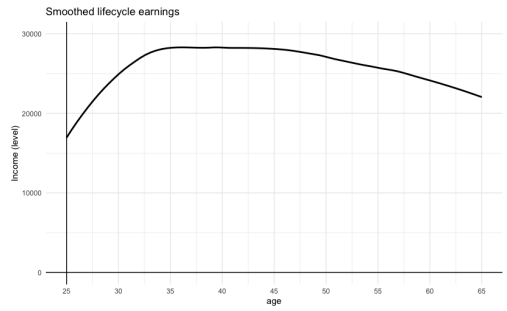


Figure 1: Deterministic life cycle component of productivity

Income and initial wealth Income is determined by the endogenous wage w and exogenous, idiosyncratic productivity $Y_{i,j}$. Productivity is split up in a deterministic lifecycle and a stochastic component. Average productivity is normalized, and together with the wage normalization adds up to an average net labor income of 26033 euros.

The lifecycle component ξ is calibrated using the DNB household survey by regressing the log of net income on age dummies and year fixed effects. The smoothened life cycle pattern of productivity is shown in figure 1. The residual of the regression is then used to calibrate the idiosyncratic, stochastic part $z_{i,j}$. Further details on the data used in calibration of the income process can be found in the appendix.

Initial wealth is a combination of a degenerate distribution at zero wealth and a log-normal distribution over non-zero support, independent of income. I calibrate the fraction of households with a zero level of wealth, p_0 as those households with no or negative financial assets in the data at age 25. With respect to the log-normal distribution, the mean is determined endogenously in order to make sure that the value of assets held by the newborns integrates up to the value bequeathed by the dying households. The variance of the log normal distribution, $\sigma^2_{a_0}$ is calculated as the variance of all the observations with a positive amount of financial assets at age 25.

[8] This price pins down the tax in percentage to price per tCO2 unit. For example a tax of 150%. 150% of 0.0023 means 0.00345 per unit of FF, which implies $0.00345 * 26033 = 89.81$ euros. One unit of FF emits 233 kCO2. Hence 4.3 units emit a ton of CO2. So the price is $89 * 4.29 = 381$ euro per tCO2

Electricity firm Similar to Hochmuth et al. (2025), I assume α_R to be 0.5. I take estimates for the elasticity of substitution from the literature, namely 1.8 for the EoS between green and brown sources in green electricity generation λ, from Papageorgiou et al. (2017) and 0.25 for the elasticity between capital and fossil fuels in brown electricity production, ω, from Coenen et al. (2024). The remaining parameter, ζ governs the capital share in the production of E_F, and is assumed to be 0.3.

Lastly, The productivity of capital in the electricity sector, for green capital A_{KR} and brown capital A_{KF} are chosen to match (1) the share of renewable electricity and (2) the relative price of electricity over fossil fuels, explained in more detail in the internal calibration section.

The results are robust to changes in ζ ranging from 0.1 to 0.5, as the parameter is relatively inconsequential due to the internal calibration of the productivity to match input shares.

Durable goods production I set the productivity level of the firm producing the brown durable good such that the price of the brown durable good equals nearly half of the mean yearly income, namely 12000 euros. I set the price gap between the green and the brown durable to 1.9, similar to Kuhn & Schlattmann (2024) who calculate this for Germany.

During the transition, I let the price premium of the green durable shrink to zero by increasing the productivity A_G in a linear fashion, such that at time T, $P_T^{d_G} = P_T^{d_G}$. In order to determine T, I extrapolate the trend of the real price decrease in electrical vehicles and heat pumps over the period 2014-2020.

Final consumption good firm I follow the literature for both the capital-share in the capital-labor composite, and the the elasticity of substitution, ν. With respect to the latter, the literature agrees on a realistically low elasticity of substitution ranging from 0.1 (Hochmuth et al., 2025; Fried et al., 2022) to 0.2 (Olovsson & Vestin, 2023). In order to be on the conservative side in terms of the tax required to decrease industry emissions, but stay within the accepted range, I set ν to 0.2.

Next, with repspect to the capital-labor share, ψ, Hassler et al. (2021) show that this parameter is unimportant in this exact problem so long as ν is not close to one. Therefore I follow Hochmuth et al. (2025) and set it to a 50-50 share. To pin down the elasticity of subsitution between the different types of fossil, ι, I employ the empirical literature estimate from Papageorgiou et al. (2017), who estimates both ι and the electricity share κ.

I internally calibrate the rates at which energy is transformed into the final good, so productivities A_{EC} and A_{FFC} in order to match the electricity input share, and the amount of fossil fuels used in industry over total fossil fuels used. Lastly, the productivity of capital-labor is set to normalize the wage to one in the initial steady state.

3.1 Internally calibrated parameters

To discipline 9 parameters that lack a clear data counterpart, I target several implied moments in the simulated data. For exact identification, I use 9 moments, which are shown in table 2. I match the model

	Moment	Calculation	Data	Model
β	Mean wealth over mean income	$\int_{i,j} a_{i,j} \;/\; \int_{i,j} wy_{i,j}$	4.2	4.48
η	Decay of wealth	$\int_i a_{i,J} \;/\; \int_i a_{i,J_W}$	0.98	0.56
$\underline{b}$	Wealth inequality at death	$\int_{i\vert_{D3}} a_{i,J} \;/\; \int_{i\vert_{D6}} a_{i,J}$	5.71	8.64
γ	Mean energy expenditure share	$\int_{i,j} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}}$	0.125	0.124
$\bar{s}$	Degree of non-homotheticity	$\int_{i,j\vert_{Q1}} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}} \;/\; \int_{i,j\vert_{Q5}} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}}$	1.6	1.63
A_{KR}	Green electricity share	$\frac{E_R}{E_R + E_F}$	0.175	0.167
A_{KF}	Relative price of electricity	$\frac{P_E}{P_{FF}}$	2.40	2.36
A_{EC}	Input share electricity final good	$\frac{E_C}{E_C + FF_C}$	0.34	0.28
A_{FFC}	Final good fossil fuel share	$\frac{FF_C}{FF_C + FF_{HH} + FF_E}$	0.46	0.48
μ_{Adj}	Share electrified 2017	$\int \phi d\mu_{2017}$	0.01	0.01
σ_{Adj}	Share electrified 2017, mid income	$\int \phi d\mu_{2017,25\%<z<75\%}$	0.0122	0.0123

Table 2: Target moments in internal calibration

implied with their data counterparts using a simulated method of moments. In doing so I choose a vector of parameters $\iota = (\beta, \eta, \underline{b}, \gamma, \bar{s}, A_{KR}, A_{KF}, A_{EC}, A_{EFF})$ in order to minimize the objective function

$$\min_{\iota} \sum_{i=1}^{9} \omega_i \left(\frac{M_i^{model}(\iota) - M_i^{data}}{M_i^{data}} \right)^2, \tag{39}$$

in which ω_i gives a weight for each moment i.

For the moments related to the wealth and income on the household side, i.e. parameters $\beta, \eta, \underline{b}$, I use Dutch micro-data. The first three moments are calculated in the data using the DNB household survey, waves 2010 - 2020. For the discount factor β, I target the ratio of mean household financial net worth (NW) over mean net household income, which in the model is normalized to 1. Second, for the strength of the bequest motive, η, I measure the mean decay of net worth in the data as the mean net worth at age 75 over the mean worth at age 65, in order to match the ages of retirement and death in the model. Over the course of 10 years, financial net worth is almost constant, decreasing slightly from 49668 to 48683 euros. Lastly, to discipline the degree of luxurity of bequests, $\underline{b}$ I target the degree of bequest inequality. I compute the mean financial net worth in the third and sixth decile at age 75, and take the ratio as the measure of inequality.

For the fourth and the fifth moment, governing the energy expenditure share of households, I use the experimental statistics on income, consumption and wealth from Eurostat for 2015. This dataset uses data from the Household Finance and Consumption Survey. By using household level data, it calculates

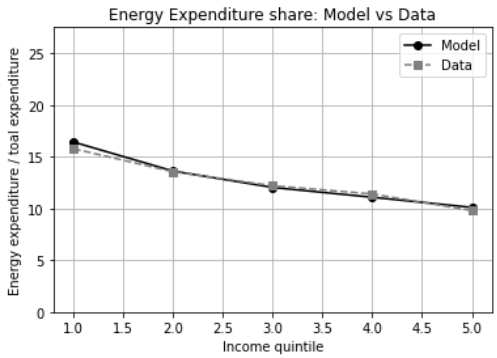


Figure 2: Energy expenditure share per income quintile in Eurostat data (2015) and in the model initial steady state

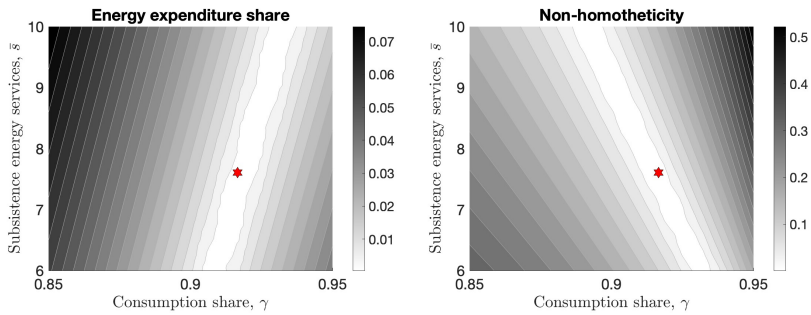


Figure 3: Absolute deviation between simulated model moment and the data target for parameters $\gamma, \bar{s}$ and A_G and their respective moments the mean energy expenditure share, degree of non-homotheticity in energy expenditures and share of households with a green durable.

expenditure shares for a variety of different goods as well as total expenditures. In figure 2 the dotted gray line shows the energy expenditure share for households in each income quintile in the data. In the internal calibration, I match the average energy expenditure share of 12.56% and the degree of non-homotheticity as measured by the energy expenditure share in the first over the last quintile, equal to 1.6. I calculate the energy expenditure share over total expenditure where the latter is defined as expenditures on all HCIP items. By matching the average, and the degree of non-homotheticity, the energy expenditure per income quintile in the model results in the black line. These two parameters together pin down the behavior targeted in the data as visualized in figure 3. They seem sufficient in getting a good fit over the entire income distribution as shown in figure 2. To discipline the productivities in the model, I use aggregate Eurostat data. I set the productivity of the green capital in electricity generation, A_{KR} to target share of electricity that is

produced from renewable sources which in Europe, 2015, is 17.5%[9]. To determine the productivity of the brown capital in electricity generation, A_{KF}, I target the relative price of electricity over the price of fossil fuels. As explained above, the price of electricity is 0.0546 per kWh of fossil energy. For the electricity price, I take the DC band price for one kWh in 2015 in Europe, which is 0.131 euros, from Eurostat [10].

The productivities for both energy sources within the production of the final consumption good, A_{EC} and A_{FFC} are disciplined by matching the input share of electricity in its energy mix and the share of fossil fuels used over total fossil fuels used. Using Eurostat aggregate data for the industry excludes the electricity sector but includes all energy used by sectors that conduct other manufacturing and industrial processes. Using natural and manufactured gas, heat, oil and petroleum products, biofossils and solid fossil fuels as a measure of total fossil fuels used in the industry, this implies 34% of all energy used is electricity. To determine the amount of fossil fuels used in the industry, I follow calculates from the energy statistics report from Eurostat[11]. Taking into account all fossil fuels used in the European economy 46% was used by the industry (defined as manufacturing, storage and services). The remainder consists of fossil fuels used in electricity production (21.8%) and by households for heating and mobility (26.2%). The remaining 6% is used for among other things mining, water supply and construction.

3.2 Baseline model validation

In order to validate the model, I evaluate the baseline switching behavior in the first years of the transition, and I compare inequality in the model and data.

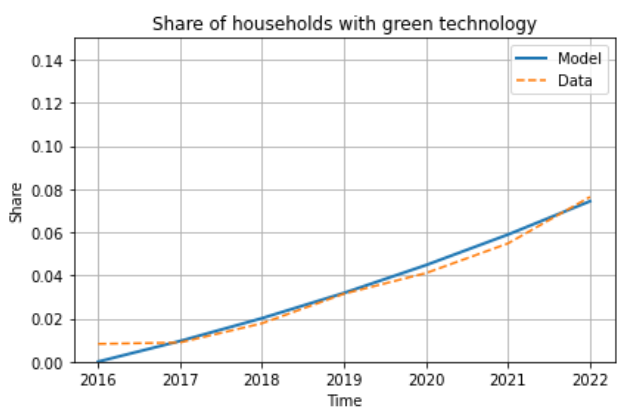


Figure 4: Baseline green technology adoption

[9]Online data code: nrg_ind_ren
[10]Online data code: nrg_pc_204
[11]Energy accounts, 2025

Baseline switching behavior I use publicly available Dutch data from RDW and CBS about the share of homeowners with a heatpump, and car owners with an electrical vehicle. The take-up of electrical vehicles is slightly slower. As in the model the good is a composite of both, I take this minimum as the share to validate against. Figure shows a good fit overall.

Moment	**EU mean** (min-max)	**Model**
Income gini	30.8 (23.7 - 37.9)	36.8
Wealth share bottom 50%	0.03 (0.0 - 0.11)	0.20
Wealth share top 10%	0.6 (0.49-0.7)	0.34

Table 3: Inequality moments

Inequality In table 3 I show some inequality statistics in the data for the European union' countries (mean, minimum and maximum) in 2015 and the model steady state. I slightly overestimate income inequality but fall within the bounds. I do heavily underestimate wealth inequality. This is caused by a precautionary savings motive for poor households, which is not so much visible in the data.

4 Policy Experiments

The economy starts in the initial steady state, and unexpectedly enters a transition. At the start of the transition, the green durable good becomes available at a price premium. The productivities of production of the green durable good A_G and green electricity A_{ER} increase over the course of the transition[12]. I compare this baseline transition case to several policy experiments where I introduce a carbon tax at the start of the transition that is phased in, also unexpected to households in steady state.

In the policy experiments, I linearly increase the carbon tax τ_c from 0 in time $t = 1$ to $\bar{\tau}_c^i$ in $T_{\tau_c}^{BL}$ time periods. I choose the exogenous amount of time periods of $T_{\tau_c}^{BL} = 25$. I then compare efficiency (GDP) and distributional welfare outcomes with respect to the baseline (no tax) case by varying the instrument, i, used to recycle the tax revenue. I consider four revenue-recycling instruments: (i) government expenditure, (ii) a uniform lump-sum redistribution, (iii) a subsidy for green electricity production, (iv) a subsidy for electrification of households[13]. I start by computing the tax rate $\bar{\tau}_c^i$ that achieves a maximum temperature increase of 2 degree celcius above the industrial level in 2065.[14] I compute the development of temperature as

[12] I extrapolate the trends in price developments over the past 10 years and assign the same growth factor for the first 15 years of the transition. For the green durable productivity I make sure it converges to the brown durable's productivity A_B.

[13] I cap the electrification for households at 40% as the revenue including the firms' tax payments exceeds the amount that can be given out as subsidies. This amount is in line with most taxes given out in Europe. In the default case, the remainder is spent on government expenditure.

[14] As the economy is not in net-zero at this point, the underlying assumption should be that some carbon capture technology keeps the temperature at 2 degree celcius.

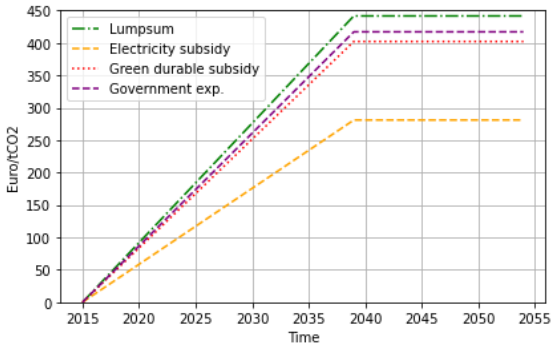


Figure 5: Carbon tax path by revenue-recycling instrument to meet the exogenous temperature target (2 degrees celcius)

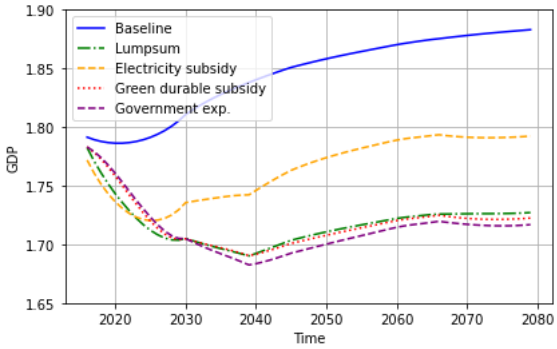
a function of total cumulative emissions, using a model discussed in appendix G.1. The figure for temperature development can be found in appendix G.1. The use of an endogenous tax rate that reaches a given temperature allows for efficiency benefits of green subsidies without explicitly modeling economic damages from temperature increases[15]. I compute the distributional welfare effects through the expenditure equivalent conditional on income and age group and I analyze how much of these welfare effects are caused by the electrification margin.

Before examining these effects, figure 5 shows the auxiliary insight into the endogenous tax path for each instrument that reaches the temperature target. Using subsidies as a recycling instrument, especially a green electricity subsidy, reduces the peak tax rate required to achieve the temperature target. The reason for the lower tax rate for subsidy recycling lies in two mechanisms. First, with a subsidy for green electricity, for a given tax rate, the electricity sector substitutes more towards renewable generated electricity. Therefore, electricity production is less emission-intensive. Second, the subsidy for renewable electricity production lowers the electricity price compared to other instruments. Therefore, for the same carbon tax, the final consumption good producer substitutes more to electricity than fossil fuels within the energy composite M.

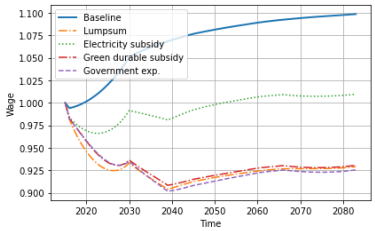
4.1 Economic activity over the transition

In order to assess the extent to which the carbon tax distorts economic activity, I assess the development of GDP and prices over the transition. The top panel of figure 6 shows the development of GDP over the transition. GDP initially declines after the climate policy introduction for all instruments because higher fossil fuel costs raise firms' production costs before the productivity of renewable capital has risen to its

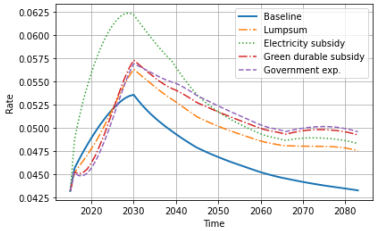
[15]Alternatively, I could set an exogenous tax rate and compute climate damages. In this case, climate damages would be lower with a subsidy rather than lump-sum redistribution or government expenditure due to lower emissions, which reduces GDP declines. In the current case I account for this by allowing for a lower tax rate under subsidy redistribution, also leading to lower GDP declines.



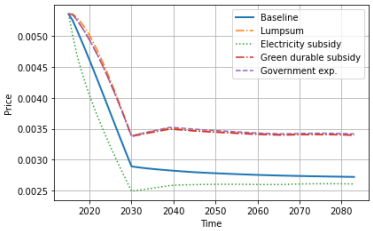
(a) GDP



(b) Wage



(c) Interest rate

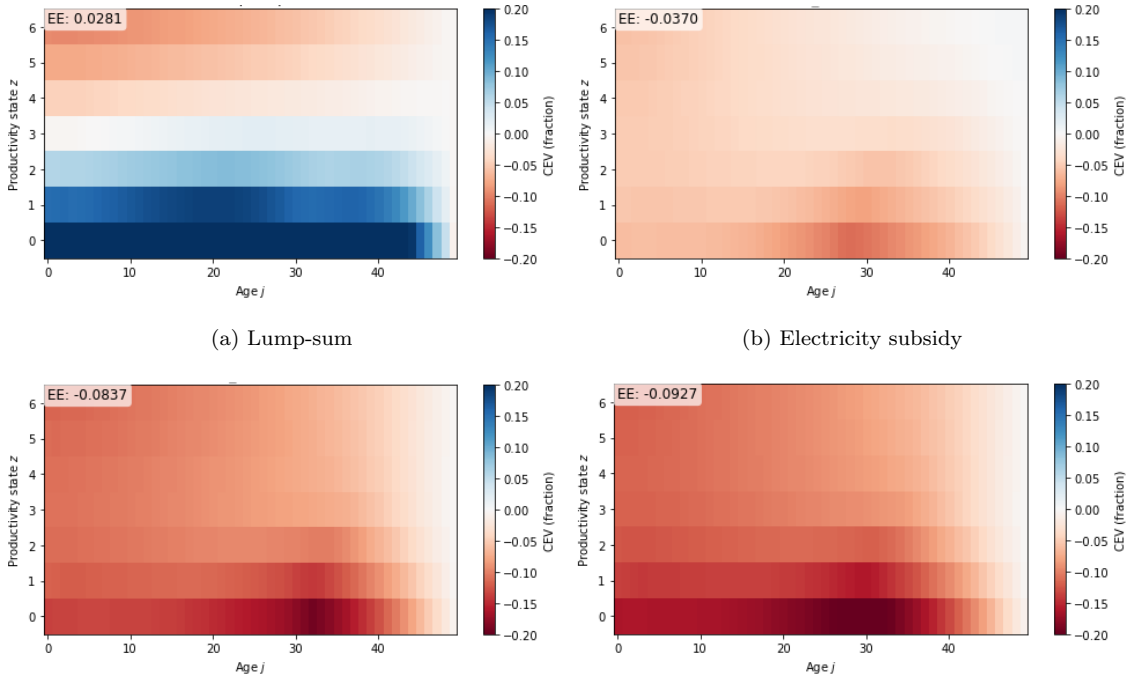


(d) Electricity price

Figure 6: GDP (top) and prices (bottom) over the course of the transition

maximum level. However, the recovery of GDP after the initial dip is highly dependent on the instrument used. In terms of reducing GDP losses stemming from the carbon tax, the electricity subsidy dominates the other recycling schemes by dampening distortions that arise from higher energy prices.

In equilibrium, the carbon tax increases the interest rate (figure 6, bottom-left) and decreases the wage (figure 6, bottom-centre). The increase in the interest rate is caused by a shift from fossil energy imports (and thus final consumption good exports) to electricity production, where the latter is more capital-intensive than the production of the final consumption good. This increase in capital demand pushes the interest rate up. This effect is amplified in the short run, as households spend their financial wealth on green technology adoption, and would otherwise have lower savings. Contrary to the interest rate, the wage decreases, because the price of the energy composite used in the production increases. As labor is only used in the final consumption good sector, under the assumption that the electricity sector is capital-intensive and does not use labor, the wage is fully determined in the final consumption good sector. The carbon tax increases the price of the energy composite, M. As the capital-labor composite is complementary to the energy composite ($\nu < 1$), the wage drops. The muted wage drop in the case of the electricity subsidy is attributable to the muted increase in the price of the energy composite M. This is due to the fact that the cost of fossil fuels is lower, following the lower endogenous tax rate estimated above, and the lower electricity price as visible in figure 6. The same reasoning holds for the muted decrease in GDP in the electricity subsidy case.



(a) Lump-sum

(b) Electricity subsidy

(c) Electrification subsidy

(d) Government spending

Figure 7: Expenditure equivalence across the age–income space for different recycling instruments. Values indicate the percentage change in expenditure in the baseline required for households to be indifferent between the baseline and policy scenario.

4.2 Distributional Welfare Effects and the Electrifcation Margin

In figure 7 I show the expenditure equivalence for different carbon tax plus redistribution instruments for the average household in a given age and productivity group for all households alive when the policy is introduced. With the exception of uniform lump-sum transfers as redistribution, all agents are worse off with a carbon tax in place. In the case of lump-sum transfers, low-income households gain from the policy due to the fact that the increase in their net disposable income, i.e. the transfers minus the wage decline, exceeds increased expenditures for poor households.

For the three other cases, government expenditure and green subsidy recycling, all agents are worse off with the climate policy in place, and poor middle-aged households suffer the largest welfare loss from a carbon tax. The electrification margin is the key driver of the high welfare loss for this group. It stems from increased expenditures, either through increased spending on the durable good when the household electrifies in order to adapt to the carbon tax and avoid high energy expenditures, or trough higher energy costs when electrification does not take place. I elaborate on these expenditures below, and why a lump-sum

transfer back to households erodes the welfare loss caused by increased expenditures for this group in the next section.

Subsidies increase welfare compared to the case of wasteful government expenditure. When the subsidy is directed at renewable electricity, the welfare losses are smaller overall compared to an electrification subsidy. The significant welfare difference between subsidy types is attributable by the attenuated negative general equilibrium effects in the electricity subsidy case. As explained above, the carbon tax and price of electricity are significantly lower following the electricity subsidy, therefore wages decrease significantly less for the same carbon tax. Overall, these two effects lead to electricity subsidies being superior over an electrification subsidy in terms of welfare effects[16]. However, the lower decrease in wages and the drop in electricity prices mostly benefit the rich, high productivity households who electrify early. On the other hand, lump-sum redistribution is progressive as it redistributes from rich households to poor.

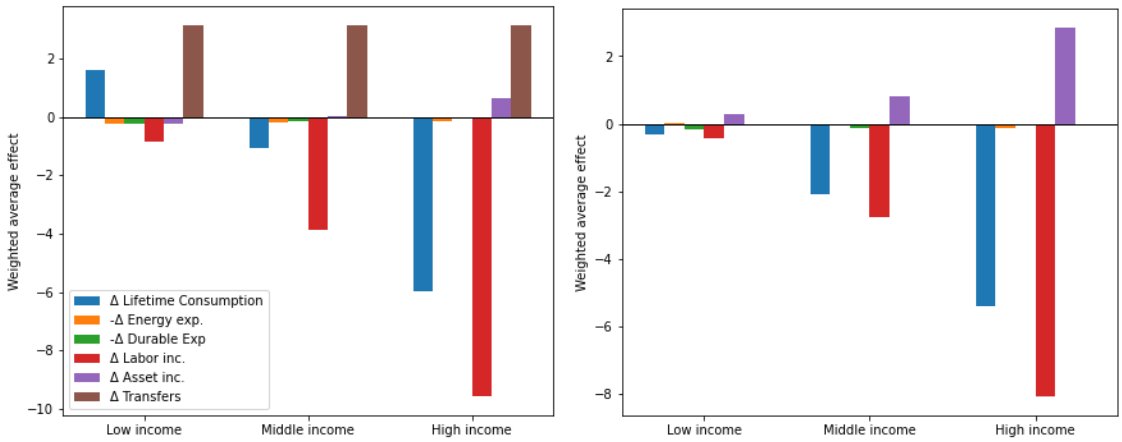
4.2.1 Mechanisms

To understand the mechanisms underlying the above distributional results, I zoom in on two sub-questions. Why is lump-sum beneficial for low-income households? And why do the poor, middle aged have the largest welfare loss without lump-sum? I answer these questions by analyzing the change in expected lifetime consumption to households after the policy introduction. The corresponding welfare effect of the consumption change depends on sign and size of the change, the timing of the change as welfare effects are discounted and the degree to which the change is spread out over time and the initial level of consumption. The latter two are caused by the concavity of the value function. Therefore, the magnitude of the change in consumption is not one-to-one translated in the magnitude of welfare effects. However, the sign of the consumption change exhibits a one-to-one relation with the expenditure equivalent. Furthermore, I can use the budget constraint to decompose changes in lifetime consumption into five different components, that I derive in appendix H.

$$\Delta\text{LC} = \Delta\text{Transfers} + \Delta\text{Labor inc.} + \Delta\text{Asset inc.} - \Delta\text{Energy exp.} - \Delta\text{Durable exp.} \tag{40}$$

Why is lump-sum beneficial for low-income households? In order to understand why lump-sum is so beneficial for poor households despite the higher tax rate on fossil fuels compared to the electricity subsidy, I plot the expected lifetime consumption change as well as the five different components for three different income groups (conditional on age) in figure 8.

[16]The combination of a green durable subsidy and electricity subsidy does not improve welfare beyond what the electricity subsidy compared to the electrification subsidy does.



(a) Lump-sum

(b) Electricity Subsidy

Figure 8: Decomposition of the change in expected lifetime consumption following policy introduction

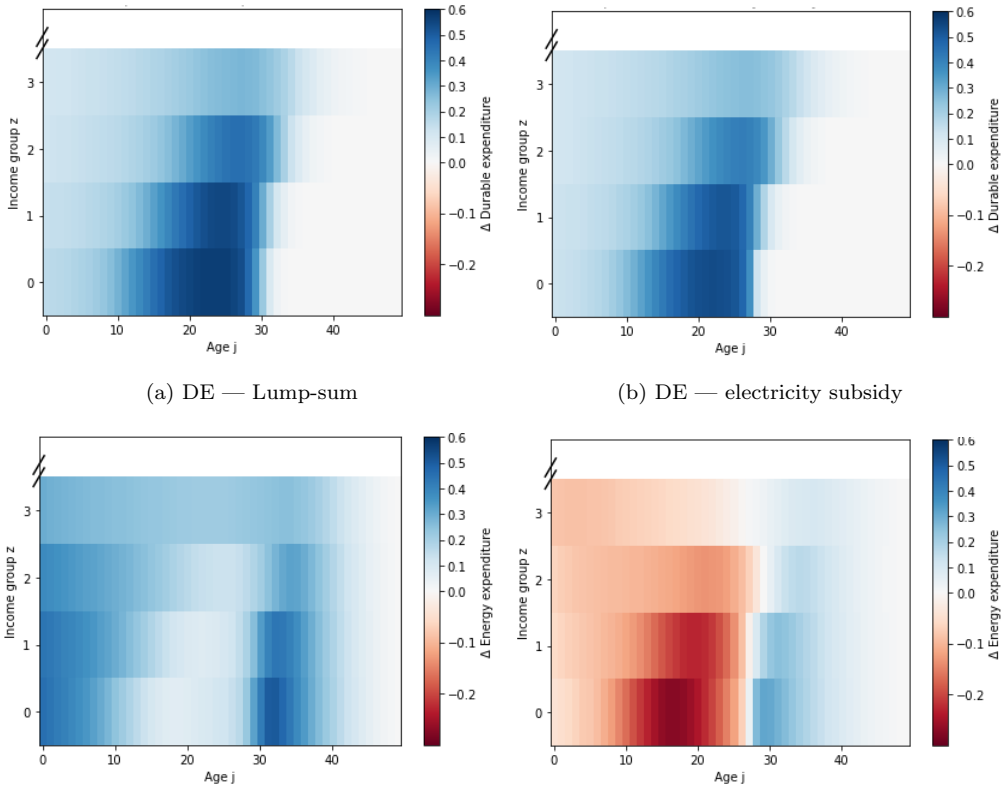
The largest part of the decrease in life-time consumption comes from the decrease in wages. Comparing within income group variation between both instruments shows that, as expected, electricity subsidies attenuate the wage decline and thus the loss in labor income. However, when focusing on between income group differences within instrument, a wage declines affects labor income of high earners by a larger magnitude. This is driven by the fact that a proportional wage decrease affects high productivity individuals more[17]. Therefore, the lump-sum transfer is actually larger in magnitude than the sum of all other negative components for low-income households as opposed to other income groups, where the lifetime consumption change is negative still. This sign difference disappears for the electricity subsidy case, where low-income households have a small negative change in expected lifetime consumption. While low-income households are worse off with electricity subsidy compared to lump-sum distribution, high-income households are actually less worse off with electricity subsidies.

Judging from figure 8, low-income households seem to loose very little consumption. However, the results in figure 8 are averaging out any life-cycle effects and hide the within-income group heterogeneity. In order to understand why especially the poorest middle-aged households in the lowest income groups are most burdened, I disaggregate the consumption response for the lowest income groups further at the age dimension.

Why do the poor, middle aged have the largest welfare loss without lump-sum? In order to explain the dark red triangle at the bottom of the heatmaps in figure 7, I analyze the heterogeneity in durable and energy expenditure because, as I will show, these make the carbon tax regressive with the existence of

[17]This is inherit to the assumption that wage decreases are uniformly distributed, which is not a necessarily standard assumption in this literature Känzig (2023)

the electrification margin. I plot the heterogeneous changes in energy expenditures and durable expenditures in figure 9 for the lowest income groups.



(a) DE — Lump-sum

(b) DE — electricity subsidy

(c) EE — Lump-sum

(d) EE — electricity subsidy

Figure 9: Heterogeneous expenditure responses across the age–income space. The top row shows changes in durable expenditures, the bottom row changes in energy expenditures. The left column corresponds to lump-sum transfers and the right column to the electricity subsidy.

It is useful to separate three groups of households: (1) the young, and rich households who would have electrified in any case, (2) the poor, old households who are not electrifying in any case, and (3) the switchers; the households who would not have electrified without the policy in place, but will do so with policy. The heterogeneous welfare burden between these groups, is caused by heterogeneous effects on lifetime consumption via expenditures on energy and the durable good.

With respect to durable expenditures the group who does not electrify in any case (group 2), is unaffected. The group who would electrify in any case (group 1) has a small negative burden, averaging to about 3000 euros in total, as they now electrify earlier on, when the green durable is more expensive. The group who is affected most, is the group of switchers who would not have electrified without climate policy but does

now (group 1). Their additional spending on the durable good averages up to 13000 euros. This is for the lowest income group people around 40-50 (j between 15-30) years old, the second lowest income group people around 45-55 years old and so on. The dark blue band in the top panels of figure 9.

Regarding energy expenditures, the impact on the different groups differs significantly in sign and magnitude between the two different instruments. With lump-sum redistribution, figure 9 panel (c), both the price of fossil fuels and the price of electricity rises. The group who never electrifies under the policy (group 2, poor, old households in right bottom corner) experiences the largest increase in energy expenditures, due to the fact that the fossil fuel costs rise the most. Within this second group, the lowest income, youngest ($j = 30$) households experience the largest increase in expenditure, due to the non-homotheticity of energy demand and the longer lifespan. The loss for them equals approximately 13.000 euros. For the group who always electrifies (group 1) the change in energy expenditures is lower in magnitude overall. Lower income households see their energy costs rise more than middle income. This is caused by two both the extensive margin, as richer households electrify earlier and the intensive margin, as rich households substitute away from energy, while poor households are closer to the subsistence requirement. The last group, the households who would not have electrified without, but will do so with policy (group 3) have unchanged energy expenditures. As they electrify with the policy and use the green technology, with a far higher energy efficiency, their energy expenditures are almost unchanged despite the increase in the electricity price, compared to using fossil fuels without policy. Second, with the electricity subsidy used as instrument in the panel (d) of figure 9. The effect on group 1, the poor, middle-aged never-electrifiers, is qualitatively similar as they see their expenditures rise, but smaller due to the lower tax rate. The effect on the switchers' and always-electrifiers' energy expenditures is negative, they decrease their energy expenditures; as the cost of electricity drops, so do their expenditures compared to the case without climate policy. This effect is largest for the switchers, as this is compared to fossil fuel energy consumption under no climate policy.

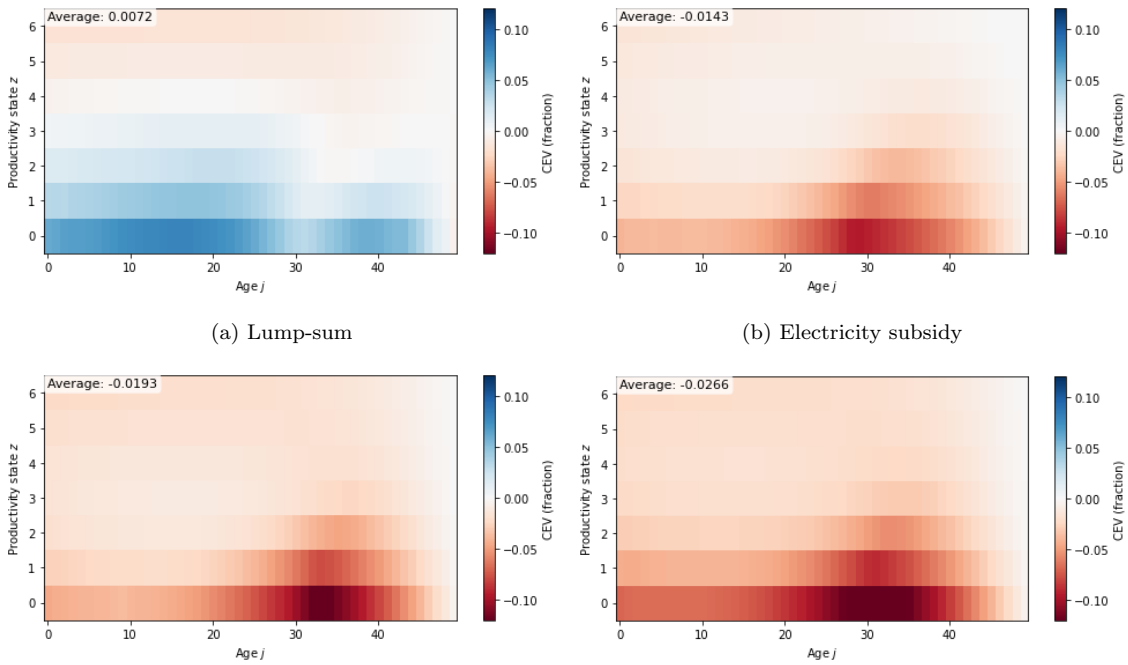
Overall, the above questions shed light on why the poorest households are most negatively affected through the electrification margin, and why this loss is neutralized with lump-sum recycling. Climate policy increases the sum of durable and energy expenditures for households that either switch to the green technology or continue using fossil fuels. These expenditure effects are more pronounced under lump-sum recycling because both fossil fuels and electricity become more expensive. In contrast, under the electricity subsidy, switchers are directly compensated through lower electricity prices, while non-switchers face a smaller increase in fossil fuel prices due to the lower tax rate. As a result, the increase in expenditures for low-income households is smaller under the subsidy than under lump-sum recycling. However, this difference in expenditure burdens is modest relative to the size of the lump-sum transfers. Under lump-sum recycling, the transfers that households receive more than offsets the additional expenditure increases faced by low-income households. To summarize, while the subsidy better targets the consumption drop arising from the electrification margin, the financial relief it provides is limited compared to lump-sum transfers.

4.2.2 Quantifying the effect of the Electrification Margin

In order to quantify the extent to which the electrification margin accounts for the regressivity of the carbon tax, I compute the differential of the expenditure equivalence above with the expenditure equivalence if all households would have been endowed with a green technology rather than a brown technology. Figure 10 plots the state-dependent difference in consumption-equivalent variation after introduction of the policy, defined as

$$\Delta CE^{Elec}(j,z) = CE_{t=1}^{\text{noElec}}(j,z) - CE_{t=1}(j,z),$$

for the gradual-tax scenario.



(a) Lump-sum

(b) Electricity subsidy

(c) Electrification subsidy

(d) Government spending

Figure 10: State-space heatmap of $\Delta CE^{Elec}(j,z)$ for different recycling instruments

In all cases except lump-sum redistribution, the electrification friction makes households significantly worse off due to climate policy. Especially, the friction makes that poor, middle-aged households suffer disproportionally from climate policy. From the discussion above, we know that the increased expenditure on durables and energy for never-electrifiers and households who electrify due to the tax are underlying these triangles. Over half of the total welfare loss of these households, is explained by the electrification margin.

On the contrary, when using lump-sum redistribution the existence of the electrification friction actually makes current, poor households a little bit better off. This is because with electrification margin, at the

start of the transition, tax revenues and therefore the lump-sum transfer is higher.

The case without electrification In contrast to the findings of Ascari et al. (2025), who show that a carbon tax combined with an electricity subsidy is regressive, I obtain the opposite result. In their framework, regressivity arises from non-homothetic energy demand and the fact that the price of energy increases even when electricity is subsidized. In my model, removing the household electrification margin eliminates the regressivity of the carbon tax. The key difference is that, unlike in their setting, the price of electricity in my economy decreases under the policy. As a result, even with non-homothetic energy demand, climate policy with revenue recycling through electricity subsidies is progressive. The reason for the declining electricity price is the assumption of continuous fuel switching of the final consumption good firm that thus also uses fossil fuel as an input. Because this firm pays the carbon tax on its fossil fuel demand, tax revenues increase substantially. These additional revenues finance the renewable electricity subsidy, and the resulting subsidy-induced reduction in electricity prices more than offsets the upward pressure from the carbon tax on fossil fuels. Consequently, electricity prices fall, leading to progressive policy effects.

4.2.3 Robustness and validity

A natural source of concern arises from the fact that the lump-sum redistribution scheme alleviates existing distortions, such as idiosyncratic risk and inequalities and is therefore welfare enhancing. In order to test whether the result found above holds true under stricter assumptions, I solve the model for an economy with the same income states but without any risk, i.e. the transition matrix is now an identity matrix, holding everything else constant. I find that both the aggregate as well as the distributional incidence of welfare losses remain qualitatively and quantitatively within the same magnitude under the assumption of no idiosyncratic risk. The aggregate gain in welfare from lump-sum distribution reduces from 2.81% to 2.47%.

This still does not speak to the concern that lump-sum might increase aggregate welfare as it alleviates current inequalities, even absent distortions such as risk. However, the comparison one can make, and the benefit to lump-sum that is clear compared to all other instruments is that lump-sum is the only instrument which is not regressive. That is, in all cases other than lump-sum the electrification margin causes poor households to suffer the greatest welfare losses as explained above. Lump-sum recycling is the only instrument that alleviates this welfare burden caused by the electrification margin for the poor.

4.3 Extension: Front-loading the carbon tax

Front-loading the carbon tax offers the advantage that the growth of the cumulative carbon stock reduces earlier, and therefore a lower tax rate can achieve the same climate target. Similarly to how $\bar{\tau}_c^i$ in $T_{\tau_c}^{BL} = 25$ was found, I find the endogenous tax rate when the tax is implemented in $T_{\tau_c}^{FL} = 15$ time periods for each instrument. In figure 11 I plot the results of the endogenous tax rates when front-loading.

The qualitative order of instruments remains the same, the main difference with the baseline case is

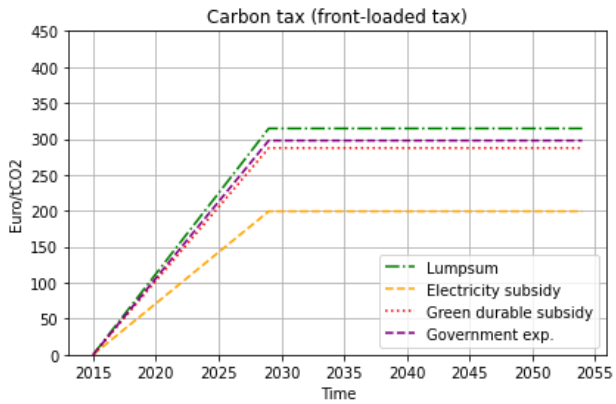


Figure 11: Carbon tax path by revenue-recycling instrument to meet the exogenous temperature target (2 degrees celcius) with front-loading

the height of the tax rates in the short versus long run. In the long run, as expected from the explained advantage above, for all recycling instruments the maximum tax rate is lower. Therefore, in the new steady state after the green transition has taken place, GDP and wages will be higher then in the baseline case. However, due to front-loading, in the short-run the carbon tax is higher then in the baseline case. And since in the short-run (up to $t = 15$) the productivity gains in the electricity sector have not yet fully materialized, there is a large short-run GDP and wage dip in the front-loading case. The latter leads to a larger negative aggregate welfare burden for generations alive when the policy is introduced. I visualize this timing trade-off for each instrument in figure 12.

The left panel shows for each instrument, the GDP decline in the new steady state (x-axis) and the aggregate current welfare loss for alive generations (y-axis). The right-hand panel shows the equivalence but with the welfare loss of the lowest income group rather than the aggregate. The solid lines show the trade-offs for the case with electrification margin, the dashed lines without electrification margin. The graph shows various results. First of all, for all instruments, the change in GDP in the new steady state compared to no policy is less negative when front-loading the tax, which is caused by the lower tax rate (figure 11 versus figure 5). What was established above but is neatly visible in this graph, is that the electrification margin affects welfare negatively, and this effect is significantly more pronounced for low-income households. Looking at the difference between front and back-loading the tax for welfare outcomes, front-loading increases the magnitude of the welfare effect: for subsidy recycling welfare becomes more negative, while with lump-sum it becomes more positive. These welfare effects are larger due to the electrification margin, especially for the poorest households. This implies that front-loading hurts the low income households the most, and this is caused by the electrification margin. Basically, as the low-income households cannot adopt green technologies (early), the higher short-run tax rate hurts these households more. Lump-sum redistribution alleviates this trade-off in the aggregate, and it vanishes even for the low-income households.

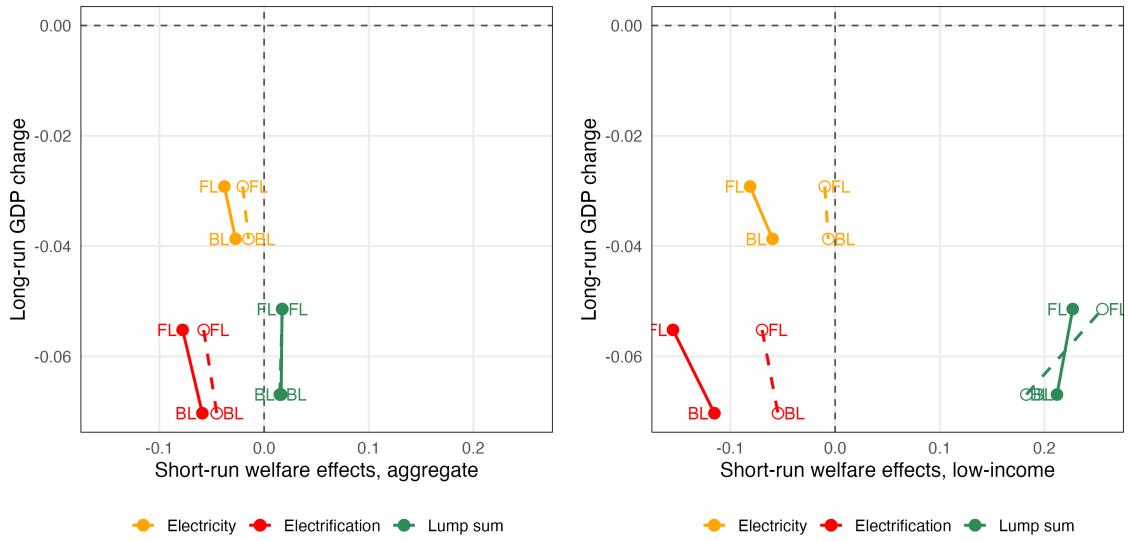


Figure 12: GDP decline in the long run versus aggregate welfare change now (left) and low-income group welfare change now (right). Solid line shows results with electrification margin, dashed line shows results without electrification margin.

5 Conclusion

This paper studied the effects of a carbon tax on efficiency and inequality in an economy containing both sides of the energy transition: electrification and decarbonization of electricity. It examined these effects under multiple revenue redistribution instruments, and different horizons at which the tax is phased in. I show that while carbon taxation robustly accelerates emission reduction, its effectiveness and welfare effects are highly dependent on the tax path and revenue recycling instrument. I also show that welfare effects are potentially unequal and that for a large part this is caused by the electrification margin.

I show that this margin causes the welfare burden to be larger at the lower-end of the income distribution. When subsidy recycling is used, these effects are salient, while they erode when households receive all tax revenues back in the form of a uniform lump-sum transfer. However, this comes at a cost of a lower GDP in the new steady state, caused by a higher maximum tax rate. This effect can however be attenuated by front-loading the tax in combination with lump-sum redistributing the money back. Under this scenario, front-loading does not decrease aggregate welfare, is even more progressive and has a lower GDP decrease than the baseline case.

Overall, the electrification margin for households created distributional effects via heterogeneous carbon intensity of energy demand, that are significant. When policy-makers are inclined on reducing emissions, it would best balance this objective with an equity objective through the use of lump-sum redistribution

rather than targetted subsidies, and phase the tax in quickly to diminish future costs of the carbon tax. These results underscore that the design of revenue recycling determines who wins and who loses from climate policy. By quantifying the relative importance of the underlying inequality channels, the paper offers guidance for policymakers seeking to balance equity and decarbonization efficiency objectives along the green transition.

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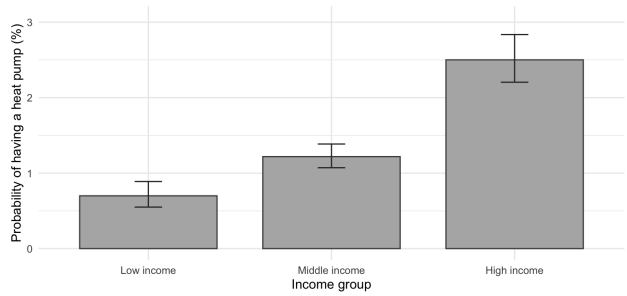

Figure 13: The predicted share of households for different income groups with a heatpump in 2018 using the WOon-dataset from Statistics Netherlands, for homeowners (37750 obs.), mean is 1.41% (dashed line). The correlation persists when controlling for square meters of the house and household age. The correlation is also found for 2021 and 2024.

Appendix

A Heatpump uptake

B Solving the household problem

The value function is given by

$$V_j(a, \phi; y, P, \mathcal{T}) = \max\{V_j^{NA}(a, \phi; y, P, \mathcal{T}), V_j^{A}(a, \phi; y, P, \mathcal{T})\} \tag{41}$$

I individually solve for the intermediate functions, which I can do for the last living period given the bequest value.

I start by solving the non-adjuster problem. I first solve for the static problem analytically. The dynamic problem then becomes

$$V_j^{NA}(a, \phi, m; y, P, \mathcal{T}) = \max_{a'} \tilde{u}(\tilde{c}) + \beta \mathbb{E}[V_{j+1}(a', \phi', m'; y, P, \mathcal{T})] \tag{42}$$

subject to

$$\tilde{w} = y + (1+r)a + \mathcal{T} - \delta_\phi - f(j)m - a' \tag{43}$$

$$e = (1-\gamma)\frac{\tilde{w}}{\phi P^E + (1-\phi)(P^{FF} + \tau_c)} + \gamma(\phi\rho + (1-\phi)\chi)\bar{s} \tag{44}$$

$$c = \gamma\big(\tilde{w} - (\phi\rho + (1-\phi)\chi)(\phi P^E + (1-\phi)(P^{FF} + \tau_c))\bar{s}\big) \tag{45}$$

$$\tilde{c} = c^{1-\gamma}\big(\frac{e}{\phi\rho + (1-\phi)\chi} - \bar{s}\big) \tag{46}$$

$$\tilde{u}(\tilde{c}) = \frac{\tilde{c}^{1-\sigma}}{1-\sigma} \tag{47}$$

$$m' = (1 - f(j) + r)m, \quad \phi' = \phi \tag{48}$$

$$a' \geq 0 \tag{49}$$

Given value function $V_j^{NA}(a, \phi, m; y, P, \mathcal{T})$ and policy functions $a'_j(a, \phi, m; y, P, \mathcal{T})$, $c_j(a, \phi, m; y, P, \mathcal{T})$, $e_j(a, \phi, m; y, P, \mathcal{T})$, I continue by solving for the adjuster value function, by recognizing that this is an interpolation of the non-adjuster problem

$$V_j^A(a, \phi; y, P, \mathcal{T}) = \max_{m'} V_j^{NA}(\tilde{a}, \phi' = 1, m'; y, P, \mathcal{T}) \tag{50}$$

$$\tilde{a} = a - (P^G - \varphi P^B) + m' \tag{51}$$

$$m' \leq \lambda P^G \tag{52}$$

Which gives policy function $\phi'_j(a, \phi, m; y, P, \mathcal{T})$,$m'_j(a, \phi, m; y, P, \mathcal{T})$ and value function $V_j^A(a, \phi, m; y, P, \mathcal{T})$

C Solving the Production side

There are three production sectors. One for the final output good, one for the two durable goods and one for electricity.

C.1 Output-good-producing firm

A representative firm produces output goods which can be used for production or converted in durable goods. Inputs are capital, labor and electricity. The technology is a constant returns to scale CES

$$Y^C(K_C, L_C, E_C, FF_C) = \Big[(1-\psi)\big(A_{KC}K_C^{\alpha_c}L_C^{1-\alpha_c}\big)^{\frac{\nu-1}{\nu}} + \psi A_{MC}M^{\frac{\nu-1}{\nu}}\Big]^{\frac{\nu}{\nu-1}} \tag{53}$$

Gives first order conditions

$$r+\delta = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} (1-\psi)\, B^{\frac{\nu-1}{\nu}-1}\, A_{KC}\alpha_c\, K_C^{\alpha_c-1} L_C^{1-\alpha_c} \tag{54}$$

$$w = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} (1-\psi)\, B^{\frac{\nu-1}{\nu}-1}\, A_{KC}(1-\alpha_c)\, K_C^{\alpha_c} L_C^{-\alpha_c} \tag{55}$$

$$P^E = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} \psi\, M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1}\right) \tag{56}$$

$$P^{FF} + \tau_c = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} \psi\, M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1-\kappa) FF_C^{\frac{\iota-1}{\iota}-1}\right) \tag{57}$$

where

$$B = A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c} \tag{58}$$

$$M = \left(\kappa E_C^{\frac{\iota-1}{\iota}} + (1-\kappa) FF_C^{\frac{\iota-1}{\iota}}\right)^{\frac{\iota}{\iota-1}}. \tag{59}$$

C.2 Durable good producers

In this economy, the durable goods can be produced by converting the consumption good (with cost 1, as its the numeraire) at some rate A_G, A_B, such that production functions are given as

$$Y^{D_G}(C_G) = A_G C_G \tag{60}$$

$$Y^{D_B}(C_B) = A_B C_B \tag{61}$$

These imply input demand functions

$$C_G = \frac{1}{A_G} Y^{D_G} \tag{62}$$

$$C_B = \frac{1}{A_B} Y^{D_B} \tag{63}$$

and prices $P^G = \frac{1}{A_G}$, $P^B = \frac{1}{A_B}$

C.3 Electricity firm

The profits of the electricity firm are given as follows

$$\begin{aligned}\Pi(K_{E_R}, K_{E_F}, FF_E) = P_E\big(\alpha_R^{\frac{1}{\lambda}} (A_R K_R)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} (A_F[\zeta K_F^{\frac{\omega-1}{\omega}} + (1-\zeta) FF^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}})^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} \\ - (r+\delta)K_{E_R} - (r+\delta)K_{E_F} - (P^{FF}+\tau_c)FF_E\end{aligned} \tag{64}$$

Equivalently, I can write the problem as follows

$$\Pi = P_E\big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} - P_{E_R} E_R - P_{E_F} E_F \tag{65}$$

subject to the technology constraints

$$E_R = A_R K_R \tag{66}$$

$$E_F = A_F[\zeta^{\frac{1}{\omega}}(A_{KF}K_F)^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} FF^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}} \tag{67}$$

Which implies I can split up the problem in two parts. First the problem where I minimize cost by choosing K_{E_R} (K_{E_R}, FF_E) subject to the constraint that production needs to equal some amount E_R (E_F). This problem gives me input demand functions for K_{E_R} (K_{E_R}, FF_E) in terms of E_R (E_F) as well as a price for E_R (E_F), namely P_{E_R} (P_{E_F}).

With these prices I can solve the problem in equation 65, to get the input demand functions for E_F and E_R in terms of some output level Y^E. Combining these with the input functions derived from the other two problems, I get input demand functions: $K_{E_R}(Y^E, r, P^F F, \tau_c)$ $K_{E_R}(Y^E, r, P^F F, \tau_c), FF_E(Y^E, r, P^F F, \tau_c)$, as functions of total output and input prices.

The first step to solving the firm problem for the supply function, is to choose E_F, E_R to minimize cost subject to the constraint of producing some level of electricity $\bar{E}$. The Lagrangian reads

$$\mathcal{L} = P_{E_R}E_R + P_{E_F}E_R + \Lambda(\bar{E} - \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}})$$

The first order conditions are

$$\frac{d\mathcal{L}}{dE_R} \implies \Lambda\alpha_R^{\frac{1}{\lambda}}(E_R)^{\frac{\lambda-1}{\lambda}-1}\big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}-1} = P_{E_R} \tag{68}$$

$$\frac{d\mathcal{L}}{dE_F} \implies \Lambda\alpha_F^{\frac{1}{\lambda}}(E_F)^{\frac{\lambda-1}{\lambda}-1}\big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}-1} = P_{E_F} \tag{69}$$

$$\frac{d\mathcal{L}}{d\Lambda} \implies \bar{E} = \big(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} \tag{70}$$

Dividing through

$$\frac{P_{E_R}}{P_{E_F}} = \frac{\alpha_R^{\frac{1}{\lambda}}(E_R)^{\frac{\lambda-1}{\lambda}-1}}{\alpha_F^{\frac{1}{\lambda}}(E_F)^{\frac{\lambda-1}{\lambda}-1}} \tag{71}$$

Solve for E_R and plug in

$$(\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F}E_F = E_R \implies \bar{E} = \big(\alpha_R^{\frac{1}{\lambda}}((\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F}E_F)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}}$$

Rewriting:

$$\bar{E} = \big(\alpha_R^{\frac{1}{\lambda}}((\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F})^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}\big)^{\frac{\lambda}{\lambda-1}} E_F \tag{72}$$

such that

$$E_F = \frac{\bar{E}}{\left(\alpha_R^{\frac{1}{\lambda}}((\frac{P_{E_R}}{P_{E_F}})^{-\lambda}\frac{\alpha_R}{\alpha_F})^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}}} \tag{73}$$

where input demand scales linearly in the amount produced $\bar{E}$, taking all prices as given.

This implies that for any output $\bar{E}$, input demand is given by

$$E_F = \alpha_F \left(\frac{P_{E_F}}{P_E}\right)^{-\lambda} \bar{E} \tag{74}$$

$$E_R = \alpha_R \left(\frac{P_{E_R}}{P_E}\right)^{-\lambda} \bar{E} \tag{75}$$

and the natural price index for E is

$$P_E = (\alpha_R P_{E_R}^{1-\lambda} + \alpha_F P_{E_F}^{1-\lambda})^{\frac{1}{1-\lambda}} \tag{76}$$

Similarly, for the intermediate electricity from renewable and fossil, from the first order conditions, prices for these are given by marginal costs

$$P_{E_R} = \frac{r+\delta}{A_R} \tag{77}$$

$$P_{E_F} = \frac{1}{A_F}(\zeta((r+\delta)/A_{KF})^{1-\omega} + (1-\zeta)(P_{FF}+\tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \tag{78}$$

and input demands are

$$K_R = \frac{1}{A_R} E_R \tag{79}$$

$$K_F = \zeta \left(\frac{r+\delta}{A_{KF} P_{E_F}}\right)^{-\omega} E_F \tag{80}$$

$$FF = (1-\zeta)\left(\frac{P_{FF}+\tau_c}{P_{E_F}}\right)^{-\omega} E_F \tag{81}$$

D Equilibrium algorithm

D.1 No tax, stationary equilibrium

Given exogenous fossil fuel price $P^F F$

1. **Initialization.**

   (a) Guess an interest rate r

(b) Compute the implied price of electricity and the prices of durables as marginal costs:

$$P^E = \left[\alpha_R \left(\frac{r}{A_R}\right)^{1-\lambda} + \alpha_F \left(\frac{1}{A_F}\left(\zeta r^{1-\omega} + (1-\zeta)(P_{FF}+\tau_c)^{1-\omega}\right)^{\frac{1}{1-\omega}}\right)^{1-\lambda}\right]^{\frac{1}{1-\lambda}} \tag{82}$$

$$P^G = \frac{1}{A_G}, \quad P^B = \frac{1}{A_B} \tag{83}$$

2. **Firm Problem.**

(a) Using a non-linear solver, solve for (K_C, E_C, FF_C, w) such that the labor market clears and the firm's first-order conditions are satisfied:

$$r + \delta = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} (1-\psi)\, B^{\frac{\nu-1}{\nu}-1}\, A_{KC}\alpha_c\, K_C^{\alpha_c-1} L_C^{1-\alpha_c} \tag{84}$$

$$w = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} (1-\psi)\, B^{\frac{\nu-1}{\nu}-1}\, A_{KC}(1-\alpha_c)\, K_C^{\alpha_c} L_C^{-\alpha_c} \tag{85}$$

$$P^E = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} \psi\, M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1}\right) \tag{86}$$

$$P^{FF} + \tau_c = \left[(1-\psi)B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}}\right]^{\frac{1}{\nu-1}} \psi\, M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1-\kappa) FF_C^{\frac{\iota-1}{\iota}-1}\right) \tag{87}$$

3. **Household Problem.**

(a) Given the price vector $P = (P^{FF}, P^G, P^B, P^E, r, w)$, compute the value and policy functions:

$$V_j^{NA}(a, \phi; Y, P, \mathcal{T}), \quad V_j^{A}(a, \phi; Y, P, \mathcal{T}),$$
$$\phi_j'(a, \phi; Y, P, \mathcal{T}), \quad a'j(a, \phi; Y, P, \mathcal{T}), \quad c_j(a, \phi; Y, P, \mathcal{T}), \quad e_j(a, \phi; Y, P, \mathcal{T})$$

(b) Compute the stationary distribution μ such that the bequest condition holds:

$$\int a'(\mathbf{x}) d\mu_J = \int \frac{a}{1+r}, d\mu - 1 \tag{88}$$

4. **Aggregation.**

(a) Compute household aggregates:

$$A_{hh} = \sum^{J-1} \int a(\mathbf{x}), d\mu_j, \qquad C_{hh} = \sum^{J-1} \int c, d\mu_j, \tag{89}$$

$$E_{hh} = \sum^{J-1} \int \phi e, d\mu_j, \qquad FF_{hh} = \sum^{J-1} \int (1-\phi) e, d\mu_j \tag{90}$$

(b) Compute net durable demands:

$$D_{hh}^G = \sum_{j=0}^{J-1} \int [\phi\delta_G + (\phi' - \phi)], d\mu_j \tag{91}$$

$$D_{hh}^B = \sum_{j=0}^{J-1} \int [(1-\phi)\delta_B - (\phi' - \phi)], d\mu_j \tag{92}$$

5. **Sectoral Output and Input Demands.**

   (a) Use market clearing for electricity:

$$Y^E = E_{hh} + E_C \tag{93}$$

   (b) Compute input demands for electricity production:

$$K_{E_R} = \frac{1}{A_R}\alpha_R\left(\frac{P^{E_R}}{P^E}\right)^{-\lambda} Y^E, \tag{94}$$

$$K_{E_F} = \zeta\left(\frac{r}{P^{E_F}}\right)^{-\omega}\alpha_F\left(\frac{P^{E_F}}{P^E}\right)^{-\lambda} Y^E \tag{95}$$

$$FF_E = (1-\zeta)\left(\frac{P^{FF}+\tau_c}{P^{E_F}}\right)^{-\omega}\alpha_F\left(\frac{P^{E_F}}{P^E}\right)^{-\lambda} Y^E \tag{96}$$

   (c) Prices of intermediate inputs:

$$P_{E_R} = \frac{r}{A_R}, \qquad P_{E_F} = \frac{1}{A_F}\left(\zeta^\omega r^{1-\omega} + (1-\zeta)^\omega (P_{FF}+\tau_c)^{1-\omega}\right)^{\frac{1}{1-\omega}} \tag{97}$$

6. **Durable Goods Sectors.**

   (a) Use market clearing for durables:

$$Y^{D_G} = D^G_{hh}, \qquad Y^{D_B} = D^B_{hh} \tag{98}$$

   (b) Compute input demands:

$$C_G = \frac{Y^{D_G}}{A_G}, \qquad C_B = \frac{Y^{D_B}}{A_B} \tag{99}$$

7. **Equilibrium Conditions.**

   (a) Check capital market clearing:

$$A = K_C + K_{E_R} + K_{E_F} \tag{100}$$

   (b) Check aggregate resource constraint:

$$Y = C_{hh} + C_B + C_G + I_{E_R} + I_{E_F} + I_C + P^{FF}(FF_{hh} + F_E) + \int \phi f(j)M - (\phi' - \phi)M', d\mu \tag{101}$$

   If either condition fails, update r and return to Step 1(b).

D.2 Transition

1. **Initialization.**

   (a) Fix a time horizon $t = 0, \ldots, T$, tolerance levels $\varepsilon_r > 0$, and a damping parameter $\eta_r \in (0, 1)$.

   (b) Choose an initial guess for the interest-rate path $\{r_t^{(i=0)}\}_{t=0}^T$.

   (c) Specify path for productivity gains $\{A_t\}_{t=0}^T$ and taxes $\{\tau_{c_t}\}_{t=0}^T$, and calculate the implied sequences for $\{P_t^E\}_{t=0}^T$ and $\{w_t\}_{t=0}^T$ from the firm problem.

2. **Outer iteration** $i = 0, 1, 2, \ldots$ (until convergence):

   (a) Compute the value function of the stationary equilibrium in the final period T, $V_j^T(a, \phi; Y, P, \mathcal{T})$. By backward induction, taking the price sequences as given, compute the value function $V_j^t(a, \phi; Y, P, \mathcal{T})$ for $t = T - 1, ..., 0$. Using the policy functions, simulate the distribution forward for T periods starting from the initial distribution μ_0, being the initial cross-sectional distribution of households μ_0 as the steady state at $t = 0$

   (b) From the distribution, for each time period t calculate aggregate capital holdings $\int a(r_i) d\mu_t$ and aggregate electricity demand from households $E_{hh} = \int \phi e d\mu_t$

   (c) **Inner iteration,** $j = 0, 1, 2, \ldots$ (until convergence): solve the static production problem to clear the market. For each time period t:

      i. Guess an interest rate r_j, calculate $P_j^E(r_j)$.

      ii. Given r_j, $P_j^E(r_j)$ calculate K_C and E_C from the first order conditions of the final consumption good firm.

      iii. Given E_{hh}, E_C and r_j calculate K^{E_R} and K_{E_F}

      iv. Check if $\int a(r_i) d\mu_t = K_C(r_j) + K^{E_R}(r_j) + K_{E_F}(r_j)$. If not, update r_j and go back to ii.

   (d) Check if $\{r_t^{(i)}\}_{t=0}^T = \{r_t^{(j)}\}_{t=0}^T$, if not update

   $$\{r_t^{(i+1)}\}_{t=0}^T = \eta_r \{r_t^{(i)}\}_{t=0}^T + (1 - \eta_r)\{r_t^{(j)}\}_{t=0}^T$$

   and continue with step 1.c.

E Deriving the aggregate resource constraint

Given production functions for durable goods, consumption and electricity

$$D_G: \quad Y^{D_G}(C_G) = A_G C_G \tag{102}$$

$$D_B: \quad Y^{D_B}(C_B) = A_B C_B \tag{103}$$

$$C: \quad Y^C(K_C, L_C, E_C, FF_C) = F^C(K_C, L_C, E_C, FF_C) \tag{104}$$

$$E: \quad Y^E(K_{E_F}, K_{E_R}, FF_E) = F^E(K_{E_F}, K_{E_R}, FF_E) \tag{105}$$

and the fact that fossil fuels are imported, the following markets need to clear[18]:

$$\text{Capital:} \quad \sum^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \tag{106}$$

$$\text{Labor:} \quad \sum^{J-1} \int y d\mu_j \ = L_C (= \bar{L}) \tag{107}$$

$$\text{Green Durable:} \quad Y^{D_G} = \sum^{J-1} \int [(\phi' - \phi) + \delta_G \phi] d\mu_j \tag{108}$$

$$\text{Brown Durable:} \quad Y^{D_B} = \sum^{J-1} \int [\delta_B (1-\phi) - \varphi(\phi' - \phi)] d\mu_j \tag{109}$$

$$\text{Fossil Fuel:} \quad FF = \sum^{J-1} \int e(1-\phi') d\mu_j + FF_E \tag{110}$$

$$\text{Electricity:} \quad Y^E = \sum^{J-1} \int e\phi' d\mu_j + E_C \tag{111}$$

the government budget constraint needs to hold

$$G + \mathcal{T} + s_{E_R} K_{E_R}(r+\delta) + s_{electr} P^G \sum_{j=0}^{J-1} \int (\phi' - \phi) d\mu_j = \tau_c [\sum_{j=0}^{J-1} \int e(1-\phi') d\mu_j + FF_E] \tag{112}$$

such that the aggregate resource constraint can be derived. Without loss of generation, I here only derive the case of lump-sump redistribution.

Starting from the household budget constraint (note that households with $\phi = 1$ are not allowed choose to switch back after electrification, such that the budget constraint can be written as):

$$\begin{aligned} wy + (1+r)a + \mathcal{T} = c + a' + \phi' e P^E + (1-\phi') e (P^{FF} + \tau_c) + \phi \delta_G P^{D_G} + (1-\phi)\delta_B P^{D_B} \\ + (\phi' - \phi)(P^{D_G} - \varphi P^{D_B}) + \underbrace{\phi f(j) M - (\phi' - \phi) M'}_{=m(\phi', \phi)} \end{aligned} \tag{113}$$

where $(\phi' - \phi)(P^{D_G} - \varphi P^{D_B})$ is the cost of electrification. Furthermore $m(\phi', \phi)$ is the loan installment payment in case of staying (having previously electrified) or receiving the loan in case of electrifying.

In the first step I integrate both sides of the budget constraint

$$\begin{aligned} wL_C + \int (1+r) a d\mu + \mathcal{T} = C_{hh} + \int a' d\mu + \int \phi' e P^E d\mu + \int (1-\phi') e (P^{FF} + \tau_c) d\mu + \int \phi \delta_G P^{D_G} d\mu + \int (1-\phi)\delta_B P^{D_B} d\mu \\ + \int (\phi' - \phi)(P^{D_G} - \varphi P^{D_B}) d\mu + \int m(\phi', \phi) d\mu \end{aligned} \tag{114}$$

[18]The conditions read as follows. Total saving must equal total capital demand. Total labor supply must equal labor demand. Production of the green durable must be equal to newly bought green durables plus depreciation (bequests don't go on the market but are passed through as the durable itself), production of brown durable equals depreciation minus what comes on the market (can be negative?). Total fossil fuel equals household usage and fossil fuels used in electricity production, lastly electricity production is equal to household demand plus final consumption good electricity input demand

Starting with the the two durable goods, in the budget constraint, rearrange terms to get these:

$$P^{D_B}\left[\int(1-\phi)\delta_B-(\phi'-\phi)d\mu\right] \quad \text{and} \quad P^{D_G}\left[\int\phi\delta_G+(\phi'-\phi)d\mu\right] \tag{115}$$

which are exactly as in market clearing conditions (7) and (8), such that I can write

$$P^{D_B}\left[\int(1-\phi)\delta_B-(\phi'-\phi)d\mu\right]=P^{D_B}Y^{D_B} \quad \text{and} \quad P^{D_G}\left[\int\phi\delta_G+(\phi'-\phi)d\mu\right]=P^{D_G}Y^{D_G} \tag{116}$$

Furthermore, plugging the production functions for $Y^{D_B}=A_GC_G$and $Y^{D_G}=A_GC_G$, and noting that in this linear technology, and with price for C equal to one, I get $P^{D_G}=\frac{1}{A_G}$ and $P^{D_B}=\frac{1}{A_B}$ these expressions simply become

$$P^{D_B}\left[\int(1-\phi)\delta_B-\varphi(\phi'-\phi)d\mu\right]=C_B \quad \text{and} \quad P^{D_G}\left[\int\phi\delta_G+(\phi'-\phi)d\mu\right]=C_G \tag{117}$$

which I can substitute in equation 114 to get

$$\begin{aligned} wL_C+\int(1+r)ad\mu+\mathcal{T}=C_{hh}+C_B+C_G+\int a'd\mu+\int\phi'eP^Ed\mu+\int(1-\phi')e(P^{FF}+\tau_c)d\mu \\ +\int m(\phi',\phi)d\mu \end{aligned} \tag{118}$$

Next, I rewrite the market clearing condition for electricity,

$$\sum^{J-1}\int e\phi'd\mu_j=Y^E-E_C \tag{119}$$

and substitute the RHS in the new aggregate budget constraint. I also take $(1+r)a$ to the right hand side, to get

$$\begin{aligned} wL_C+\mathcal{T}=C_{hh}+C_B+C_G+\int[a'-(1+r)a]d\mu+(Y^E-E_C)P^E+\int(1-\phi')e(P^{FF}+\tau_c)d\mu \\ +\int m(\phi',\phi)d\mu \end{aligned} \tag{120}$$

Using the capital market clearing condition, I rewrite $\int[a'-(1+r)a]d\mu$ as follows:

$$[a'-(1+r)a]d\mu=K_{t+1}-(1+r)K_t \tag{121}$$

Noting that $K_t=K_{C,t}+K_{E_R,t}+K_{E_F,t}$, I split up aggregate capital to get:

$$K_{t+1}-(1+r)K_t=K_{C,t+1}-K_{C,t}+K_{E_R,t+1}-K_{E_R,t}+K_{E_F,t+1}-K_{E_F,t}-rK_{C,t}-rK_{E_R,t}-rK_{E_F,t} \tag{122}$$

$$=I_{C,t}+I_{E_R,t}+I_{E_F,t}-rK_{C,t}-rK_{E_R,t}-rK_{E_F,t} \tag{123}$$

which I substitute in the aggregate budget constraint

$$wL_C + \mathcal{T} = C_{hh} + C_B + C_G + I_C + I_{E_R} + I_{E_F} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} + (Y^E - E_C)P^E + \int (1-\phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \tag{124}$$

Take $-rK_{C,t}$ and $-P^E E_C$ to the other side to get

$$wL_C + rK_{C,t} + P^E E_C + \mathcal{T} = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} - rK_{E_R} - rK_{E_F} + Y^E P^E + \int (1-\phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \tag{125}$$

and since standard FOC for the final production good imply that marginal products are equalized with input prices, Eulers theorem implies that $Y^C = wL_C + (\delta + r)K_{C,t} + P^E E_C + (P^{FF} + \tau_c)FF_C$:

$$Y^C + \mathcal{T} = (P^{FF} + \tau_c)FF_C + C_{hh} + C_B + C_G + I_C + I_{E_R} + I_{E_F} - rK_{E_R} - rK_{E_F} + Y^E P^E + \delta K_C + \int (1-\phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \tag{126}$$

There is a zero-profit condition in the market for electricity which reads that

$$Y^E P^E - (\delta + r)K_{E_R,t} - (\delta + r)K_{E_F,t} - FF_E(P^{FF} + \tau_c) = 0 \tag{127}$$

So I can substitute for $Y^E P^E - rK_{E_R,t} - rK_{E_F,t}$ in the aggregate budget constraint

$$Y^C + \mathcal{T} = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + (P^{FF} + \tau_c)FF_C + FF_E(P^{FF} + \tau_c) + \delta(K_C + K_{ER} + K_{EF}) + \int (1-\phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \tag{128}$$

Next I impose the government budget constraint such that $\mathcal{T}$ cancels with $\tau_c \int e(1-\phi')d\mu + \tau_c FF_E + \tau_c FF_C$

$$Y^C = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + P^{FF}FF_C + FF_E P^{FF} + \delta(K_C + K_{ER} + K_{EF}) + P^{FF}\int (1-\phi')ed\mu + \int m(\phi', \phi)d\mu \tag{129}$$

Impose market clearing for fossil fuels (with abroad) $FF = FF_C + FF_E + \int (1-\phi')ed\mu$

$$Y^C = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + P^{FF}FF + \int m(\phi', \phi)d\mu + \delta(K_C + K_{ER} + K_{EF}) \tag{130}$$

Substitute in

$$\int m(\phi', \phi)d\mu = \int \phi f(j)M - (\phi' - \phi)M'd\mu \tag{131}$$

And rewrite capital investment

$$Y_t^C = C_{hh,t} + C_{B,t} + C_{G,t} + K_{t+1} - K_t + P^{FF}FF_t + \int \phi f(j)M_t - (\phi' - \phi)M_{t+1}]d\mu \tag{132}$$

Which reads as follows. Total production can either be used for consumption of the household directly, can be converted in brown or durable goods, can be invested in capital, can be spend on fossil fuels or spend on repaying the loan. In steady state,

$$Y^C = C_{hh} + C_B + C_G + \delta K + P^{FF} FF + M(r + r_m) \tag{133}$$

F Calibration data

DNB household survey

The DNB household survey consists of longtidunal data following about 2000 households. The survey contains multiple topics, such as work, pensions, mortgages, income, assets, liabilities and others. I use information about income, assets and liabilities from the AGI (Aggregated data on income) and AGW (Aggregated data on assets, liabilities and mortgages) datasets. These datasets contain information about for example total assets or income, as a sum of individual components that were asked in the surveys. The data I use is from the years 2010 to 2020.

To estimate the income process, I constrain the sample to the ages 25-65. I delete observations with a negative net labor income or net labor income that exceeds 300.000 euros per year, to not include outliers. For the wealth moments, I treat negative wealth levels as zero-wealth in order to match the models no unsecured borrowing constraint. Net worth is calculated as the sum of checking account, savings, bond holdings, stock holdings and mutual funds, and non-mortgage debt is subtracted.

G Climate Model

G.1 The Climate Model

The climate module is exactly taken from Douenne et al. (2025). The temperature is given as

$$Z_{t+1} = Z_t + \epsilon \left(\zeta \mathcal{E}_t - Z_t \right), \tag{134}$$

where cumulative emissions evolve according to

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}. \tag{135}$$

I set Z_0, ϵ, ζ, $\mathcal{E}_0$, and the exogenous emissions path $\{E_t^{\text{ex}}\}_{t \geq 0}$ externally, and take all values from Douenne et al. (2025).

G.2 Mapping model outcomes to emissions

In the model, I explicitly track fossil fuel demand, FF_t, as the sum of direct household and firm fossil fuel demand and indirect fossil fuel demand in electricity production:

$$FF_t = FF_{E,t} + FF_{C,t} + FF_{HH,t}. \tag{136}$$

The exogenous price of fossil fuels is calibrated as the price of 1000 kWh of energy, where 67% stems from gasoline and 33% from natural gas. The 670 kWh gasoline component corresponds to

$$\frac{670}{8.9} = 75.2 \text{ liters of gasoline}, \tag{137}$$

and the 330 kWh natural gas component corresponds to

$$\frac{330}{10} = 33 \text{ m}^3 \text{ of natural gas}. \tag{138}$$

Using emission factors of 2.3 kg CO_2 per liter of gasoline and 1.9 kg CO_2 per m^3 of natural gas, one unit of fossil fuel in the model produces

$$75.2 \cdot 2.3 + 33 \cdot 1.9 = 235.66 \text{ kg } CO_2. \tag{139}$$

Hence, model emissions implied by fossil fuel demand are given by

$$E_t^M = 235.66 \cdot FF_t, \tag{140}$$

with E_t^M measured in kg CO_2.

G.3 Temperature outcomes

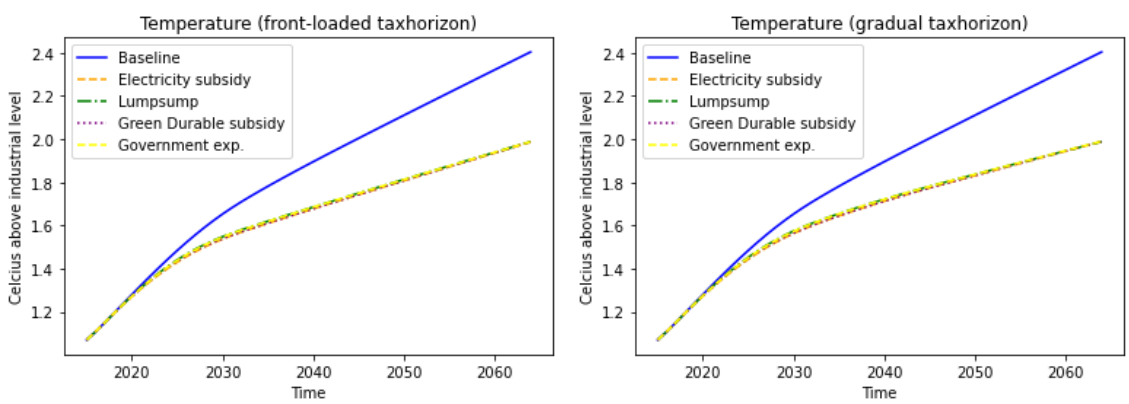

Figure 14: Temperature paths by implementation horizon and revenue-recycling instrument in order to meet the exogenous temperature target (2 degrees celcius)

H Lifetime Consumption Decomposition

I derive the decomposition of lifetime consumption. In doing so, I express the total expected lifetime consumption (plus savings bequests) of a household as the sum of its lifetime income sources, net of energy and durable expenditures.

Step 1: Per-period budget constraint. At age j and time t, the household's budget constraint is:

$$c_{j,t} + P^{\phi_{j,t}} e_{j,t} + a'_{j,t} = \mathcal{T}_t + w_t y_j + (1 + r_t) a_{j,t} - \delta_{\phi_{j,t}} - f(j) m_{j,t} + (\phi'_{j,t} - \phi_{j,t}) \left[m'_{j,t} - P_t^G + \varphi P^B \right] \quad (141)$$

where the left-hand side contains consumption, energy expenditure, and savings, and the right-hand side contains transfers, labor income, asset income, maintenance costs, loan repayments, and the net cost of switching to a green durable. To keep notation compact, define:

$$\mathrm{EE}_{j,t} \equiv P^{\phi_{j,t}} e_{j,t} \quad (142)$$

$$\mathrm{DE}_{j,t} \equiv \delta_{\phi_{j,t}} + f(j) m_{j,t} - (\phi'_{j,t} - \phi_{j,t}) \left[m'_{j,t} - P_t^G + \varphi P^B \right] \quad (143)$$

as energy expenditure and net durable expenditure respectively. The budget constraint then simplifies to:

$$c_{j,t} = \mathcal{T}_t + w_t y_j + (1 + r_t) a_{j,t} - a'_{j,t} - \mathrm{EE}_{j,t} - \mathrm{DE}_{j,t} \quad (144)$$

Step 2: Summing over the remaining lifetime. Consider a household of age j_0 at the time the policy is introduced, $t = 0$. This household has $S \equiv J - 1 - j_0$ remaining periods of life. Summing equation (144) over $s = 0, 1, \ldots, S$, where time $t = s$ and the household's age at time s is $j_0 + s$:

$$\sum_{s=0}^{S} c_{j_0+s,\,s} = \sum_{s=0}^{S} \left[\mathcal{T}_s + w_s y_{j_0+s} + (1 + r_s) a_{j_0+s,\,s} - a'_{j_0+s,\,s} - \mathrm{EE}_{j_0+s,\,s} - \mathrm{DE}_{j_0+s,\,s} \right] \quad (145)$$

Step 3: Telescoping the asset terms. Using the fact that $a'_{j,t} = a_{j+1,\,t+1}$, the asset terms form a telescoping sum. Writing them out explicitly:

$$\begin{aligned} \sum_{s=0}^{S} \left[(1 + r_s) a_{j_0+s,\,s} - a'_{j_0+s,\,s} \right] = {} & \left[(1 + r_0) a_{j_0,0} - a_{j_0+1,\,1} \right] \\ & + \left[(1 + r_1) a_{j_0+1,\,1} - a_{j_0+2,\,2} \right] \\ & + \cdots \\ & + \left[(1 + r_S) a_{j_0+S,\,S} - a_{j_0+S+1,\,S+1} \right] \end{aligned}$$

For each $s \geq 1$, the asset $a_{j_0+s,\,s}$ appears as $-a_{j_0+s,\,s}$ from the previous row and $+(1 + r_s) a_{j_0+s,\,s}$ from its own row, netting to $+r_s \cdot a_{j_0+s,\,s}$. The final term $a_{j_0+S+1,\,S+1} = a_{J,T}$ is the stock of assets left at death, which I denote as the bequest a_J. The telescoping sum therefore equals:

$$\sum_{s=0}^{S} \left[(1 + r_s) a_{j_0+s,\,s} - a'_{j_0+s,\,s} \right] = (1 + r_0)\, a_{j_0,0} + \sum_{s=1}^{S} r_s\, a_{j_0+s,\,s} - a_J \quad (146)$$

Step 4: Lifetime consumption identity. Substituting (146) back into (145) and moving the bequest a_J to the left-hand side:

$$\underbrace{\sum_{s=0}^{S} c_{j_0+s,\,s} + a_J}_{\text{Lifetime Consumption}} = \underbrace{(1+r_0)\,a_{j_0,0}}_{\text{initial wealth}} + \underbrace{\sum_{s=0}^{S} w_s\, y_{j_0+s}}_{\text{labor income}} + \underbrace{\sum_{s=1}^{S} r_s\, a_{j_0+s,\,s}}_{\text{asset income}} + \underbrace{\sum_{s=0}^{S} \mathcal{T}_s}_{\text{transfers}} - \underbrace{\sum_{s=0}^{S} \mathrm{EE}_{j_0+s,\,s}}_{\text{energy exp.}} - \underbrace{\sum_{s=0}^{S} \mathrm{DE}_{j_0+s,\,s}}_{\text{durable exp.}} \tag{147}$$

This identity states that total lifetime consumption (including bequests) equals initial wealth plus the present value of all income flows, net of energy and durable expenditures. Initial wealth $(1+r_0)a_{j_0,0}$ is identical across the policy and no-policy case at $t=0$, since the policy is introduced unexpectedly.

Step 5: Change in lifetime consumption. Taking the difference $\Delta(\cdot)$ between the policy scenario and the no-policy baseline, and noting that initial wealth cancels:

$$\begin{aligned}\Delta\left[\sum_{s=0}^{S} c_{j_0+s,\,s} + a_J\right] &= \Delta\sum_{s=0}^{S}\mathcal{T}_s + \Delta\sum_{s=0}^{S} w_s\, y_{j_0+s} + \Delta\sum_{s=1}^{S} r_s\, a_{j_0+s,\,s} \\ &\quad - \Delta\sum_{s=0}^{S}\mathrm{EE}_{j_0+s,\,s} - \Delta\sum_{s=0}^{S}\mathrm{DE}_{j_0+s,\,s}\end{aligned} \tag{148}$$

which we write compactly as:

$$\boxed{\Delta\,\mathrm{LC} = \Delta\,\text{Transfers} + \Delta\,\text{Labor income} + \Delta\,\text{Asset income} - \Delta\,\text{Energy expenditure} - \Delta\,\text{Durable expenditure}} \tag{149}$$

where $\Delta\,\mathrm{LC}$ denotes the change in lifetime consumption relative to the baseline. A positive value of $\Delta\,\mathrm{LC}$ means the policy increases the household's total lifetime consumption capacity. This decomposition allows us to isolate the channels through which climate policy affects household welfare: general equilibrium effects on wages and asset returns, direct redistribution through transfers, and expenditure effects arising from the electrification margin.